



卷积神经网络-2

Convolutional Neural Network (CNN)-2



- 01 卷积基础网络(CNN Backbones)
- 02 网络可视化(CNN Visualization)
- 03 预训练与微调(Pretraining and Finetuning)
- 04 小结 (Summary)



PART ONE

卷积基础网络

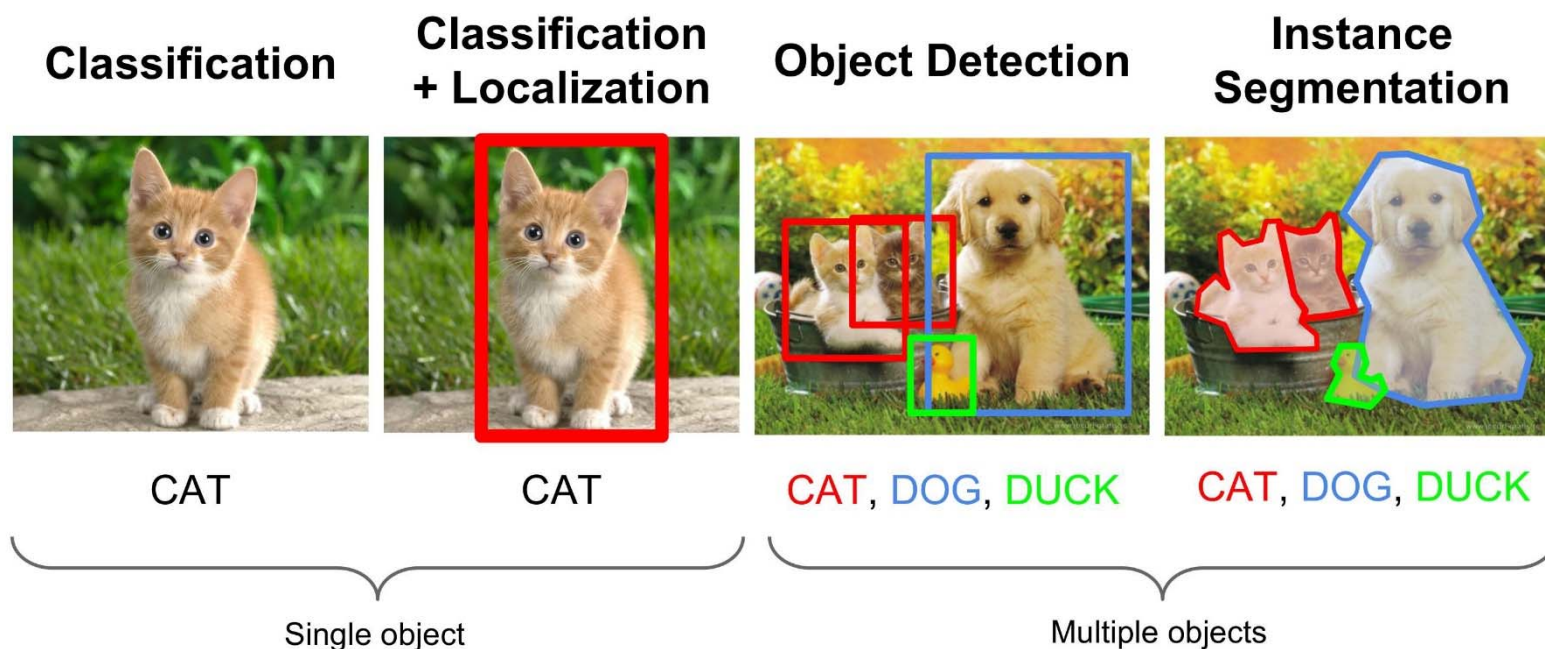
CNN Backbones



基础网络 (Backbones)

➤ ILSVRC: ImageNet Large Scal Visual Recognition Challenge

图像分类比赛ILSVRC是最重要的计算机视觉挑战赛之一，由斯坦福大学举办。比赛所用ImageNet数据集中1.2M张图像，共1000个类别



基础网络 (Backbones)

■ 典型的卷积神经网络

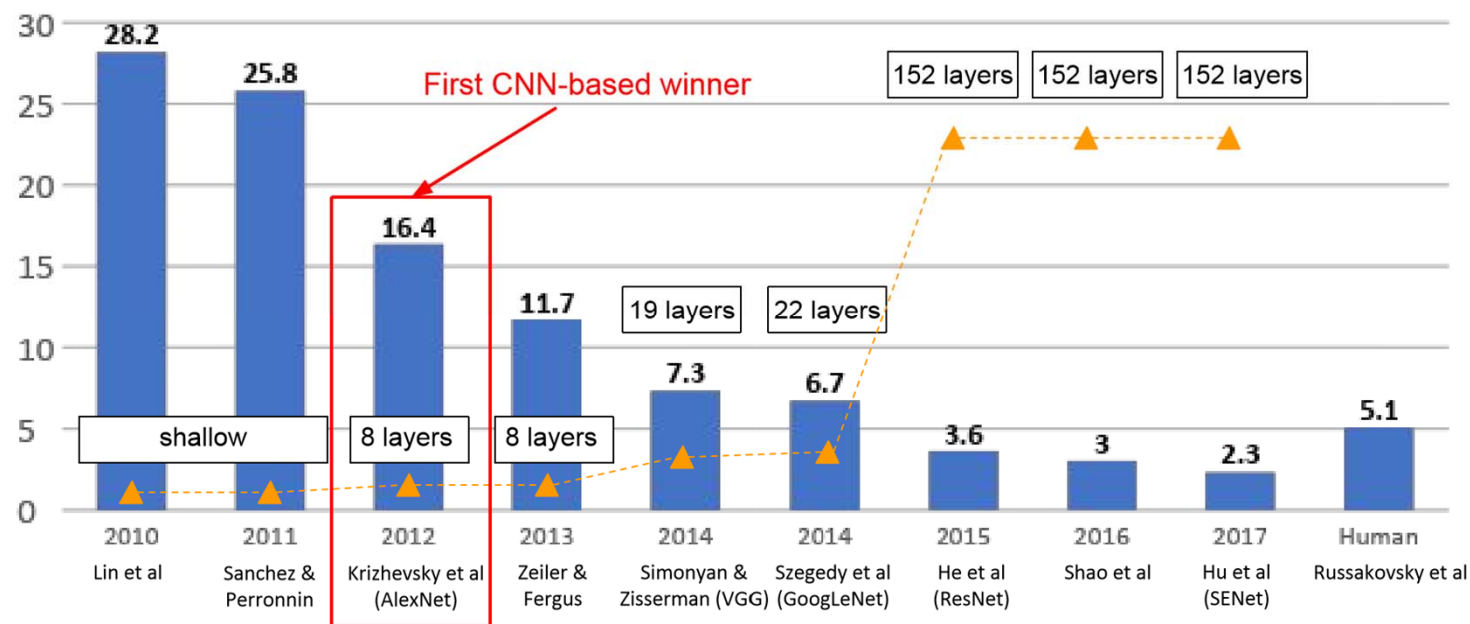
➤ AlexNet

➤ VGG

➤ GoogLeNet

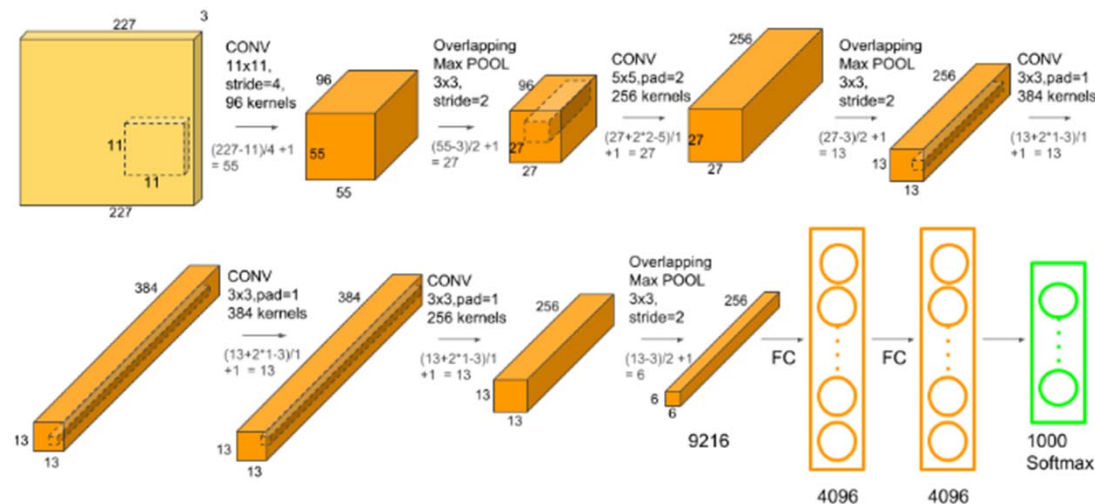
➤ ResNet

➤ DenseNet



■ AlexNet

- 2012 ILSVRC winner
 - (top 5 error of 16% compared to runner-up with 26% error)
- 第一个现代深度卷积神经网络模型
- 采用ReLU作为非线性激活函数，使用Dropout防止过拟合
- 5个卷积层、3个池化层和3个全连接层

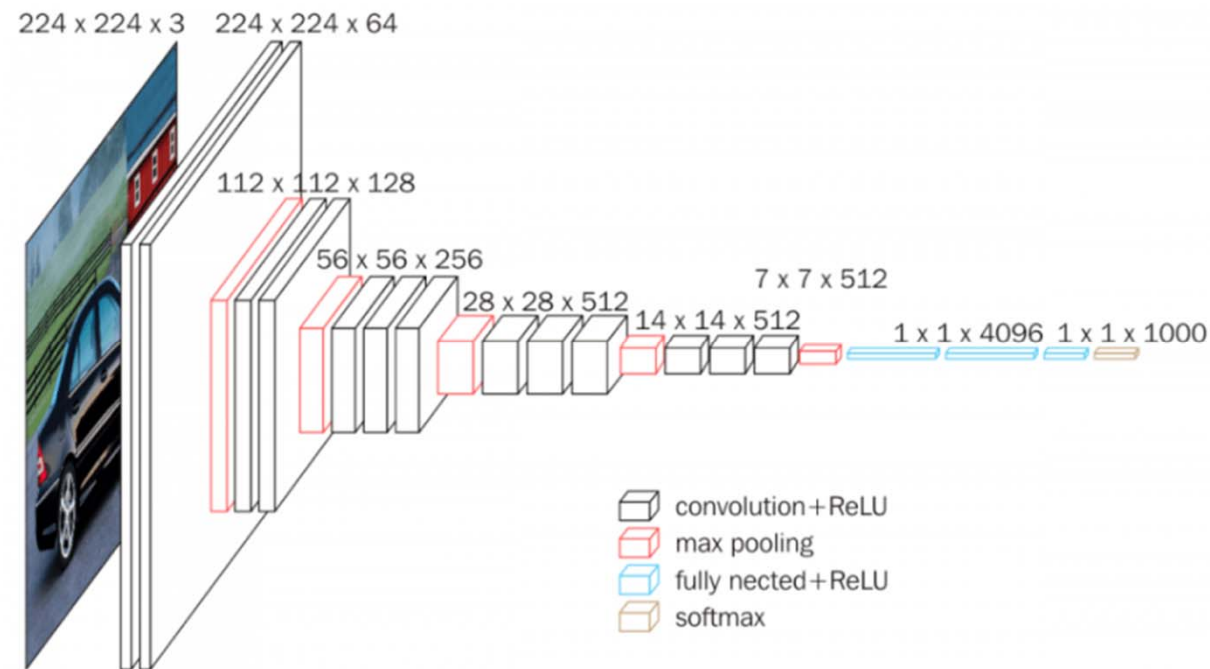


Krizhevsky *et al.*, ImageNet Classification with Deep Convolutional Neural Networks. NIPS2012.



基础网络 (Backbones)

■ VGG--牛津大学视觉几何组



Simonyan *et al.*, Very Deep Convolutional Networks for Large-Scale Image Recognition, ICLR2015.



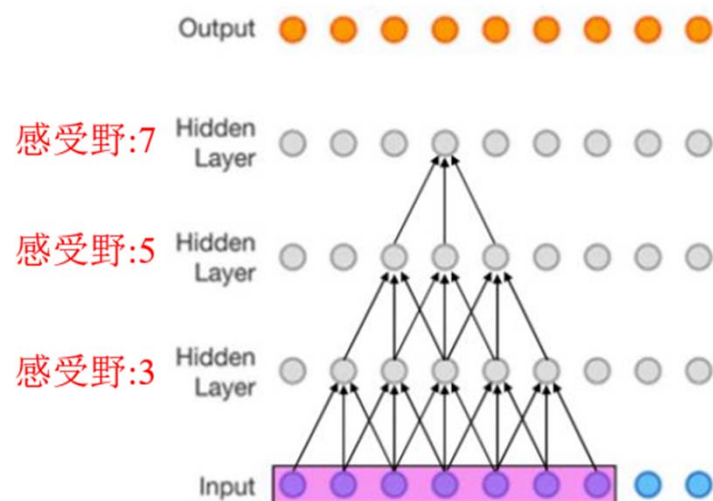
■ VGG--牛津大学视觉几何组

➤ VGG采用连续多个 3×3 的卷积核代替AlexNet中的较大卷积核:

--使用3个 3×3 卷积核来代替 7×7 卷积核

--使用2个 3×3 卷积核来代替 5×5 卷积核

- 感受野



- 参数量

输入和输出的通道数均为 C

- 7×7 滤波器, 参数量 $49C^2$

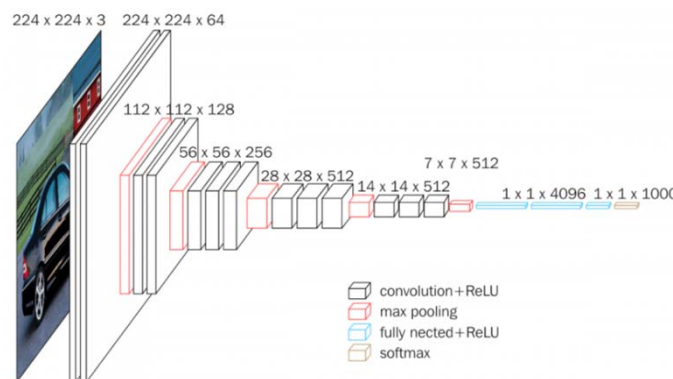
- 连续3次 3×3 的滤波器, 参数量 $9C^2 \times 3$

Simonyan *et al.*, Very Deep Convolutional Networks for Large-Scale Image Recognition, ICLR2015.



■ VGG--牛津大学视觉几何组

- VGG采用连续多个 3×3 的卷积核代替AlexNet中的较大卷积核：
 - 使用3个 3×3 卷积核来代替 7×7 卷积核
 - 使用2个 3×3 卷积核来代替 5×5 卷积核
- 优点：在保证具有相同感知野的条件下，减小网络参数量，增大网络深度，提升神经网络性能



Simonyan *et al.*, Very Deep Convolutional Networks for Large-Scale Image Recognition, ICLR2015.



基础网络 (Backbones)

■ VGG

ConvNet Configuration					
A	A-LRN	B	C	D	E
11 weight layers	11 weight layers	13 weight layers	16 weight layers	16 weight layers	19 weight layers
input (224 × 224 RGB image)					
conv3-64	conv3-64 LRN	conv3-64 conv3-64	conv3-64 conv3-64	conv3-64 conv3-64	conv3-64 conv3-64
maxpool					
conv3-128	conv3-128	conv3-128 conv3-128	conv3-128 conv3-128	conv3-128 conv3-128	conv3-128 conv3-128
maxpool					
conv3-256 conv3-256	conv3-256 conv3-256	conv3-256 conv3-256	conv3-256 conv3-256 conv1-256	conv3-256 conv3-256 conv3-256	conv3-256 conv3-256 conv3-256 conv3-256
maxpool					
conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512 conv1-512	conv3-512 conv3-512 conv3-512	conv3-512 conv3-512 conv3-512 conv3-512
maxpool					
conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512 conv1-512	conv3-512 conv3-512 conv3-512	conv3-512 conv3-512 conv3-512 conv3-512
maxpool					
FC-4096					
FC-4096					
FC-1000					
soft-max					

- 滤波器的卷积核大小均为 3×3 , stride 1, pad 1
- 3×3 Max pooling, stride 2



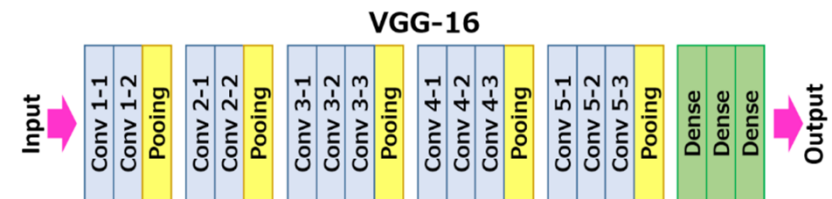
$$(227-11)/4+1 = 55$$

基础网络 (Backbones)

■ VGG

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maxpool					
conv3-128	conv3-128	conv3-128 conv3-128	conv3-128 conv3-128	conv3-128 conv3-128	conv3-128 conv3-128
maxpool					
conv3-256 conv3-256	conv3-256 conv3-256	conv3-256 conv3-256 conv3-256	conv3-256 conv3-256 conv3-256	conv3-256 conv3-256 conv3-256	conv3-256 conv3-256 conv3-256 conv3-256
maxpool					
conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512 conv3-512	conv3-512 conv3-512 conv3-512	conv3-512 conv3-512 conv3-512 conv3-512
maxpool					
conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512 conv3-512	conv3-512 conv3-512 conv3-512	conv3-512 conv3-512 conv3-512 conv3-512
maxpool					
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FC-4096					
FC-1000					
soft-max					

- 滤波器的卷积核大小均为 3×3 , stride 1, pad 1
- 3×3 Max pooling, stride 2



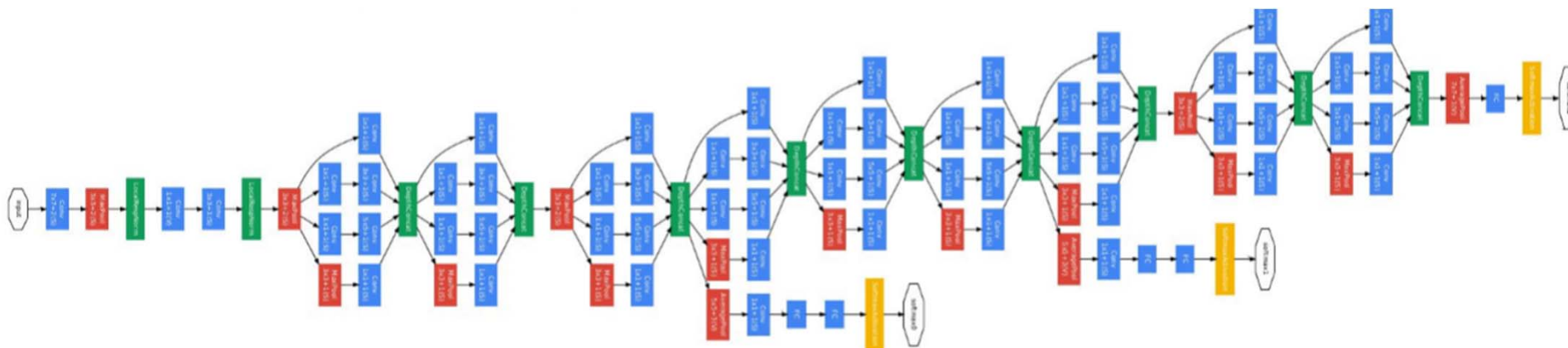
7.3% top 5 error in 2014 ILSVRC

定位任务第一名，分类任务第二名



■ GoogLeNet

- 2014 ILSVRC winner (错误率: 6.7%)
- 22层, 参数量 4M (vs AlexNet: 8层, 参数量60M, 错误率16.4%)
- 由多个Inception模块堆叠而成
- 没有全连接层



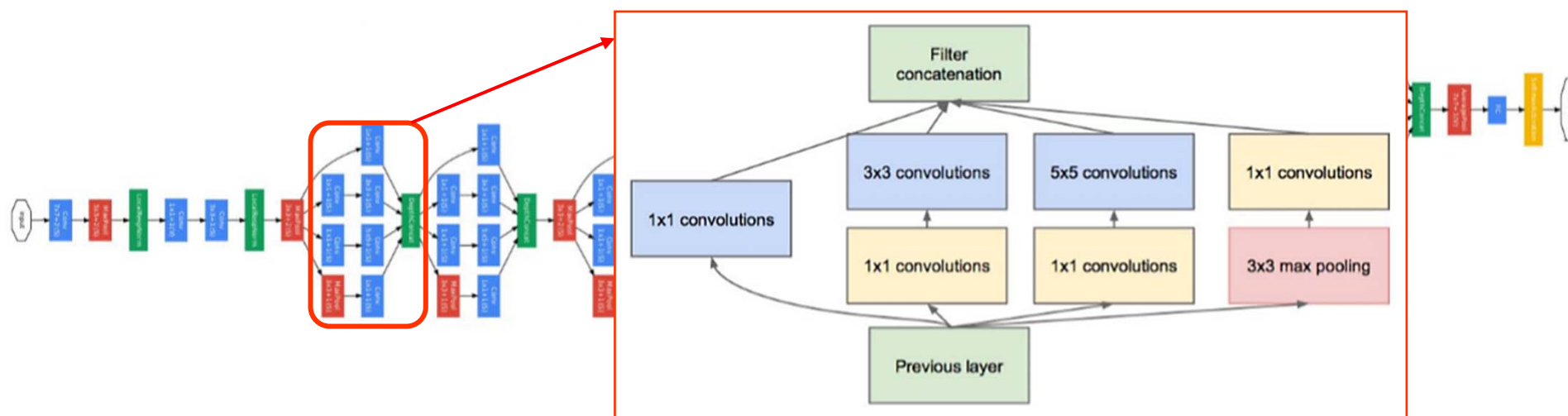
Szegedy et al., Going Deeper with Convolutions, CVPR2015.



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Inception 模块



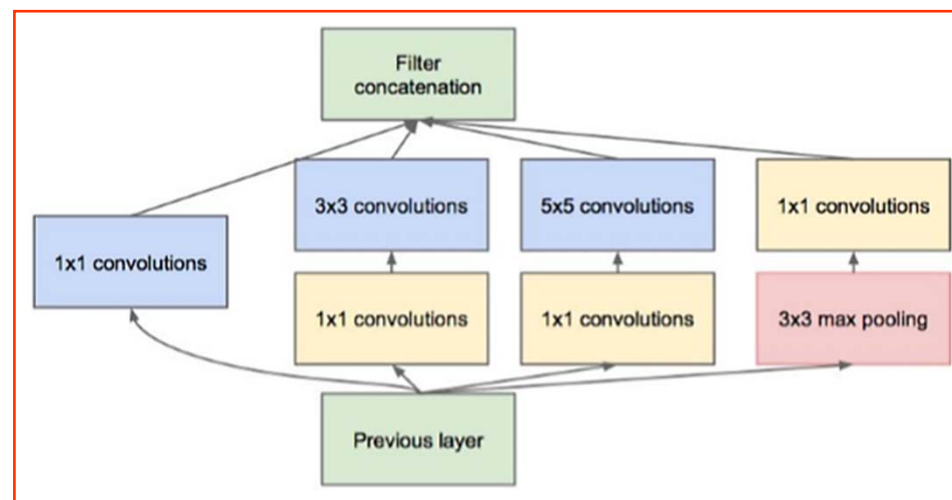
Szegedy et al., Going Deeper with Convolutions, CVPR2015.



■ GoogLeNet-- Inception模块

- 同时使用多个不同大小的卷积操作 (即 1×1 、 3×3 、 5×5 等卷积核), 并将得到的特征在通道维度上拼接(Concatenation)作为最终的输出
- 采用不同大小的卷积核意味着不同大小的感受野, 最后拼接意味着不同尺度特征的融合

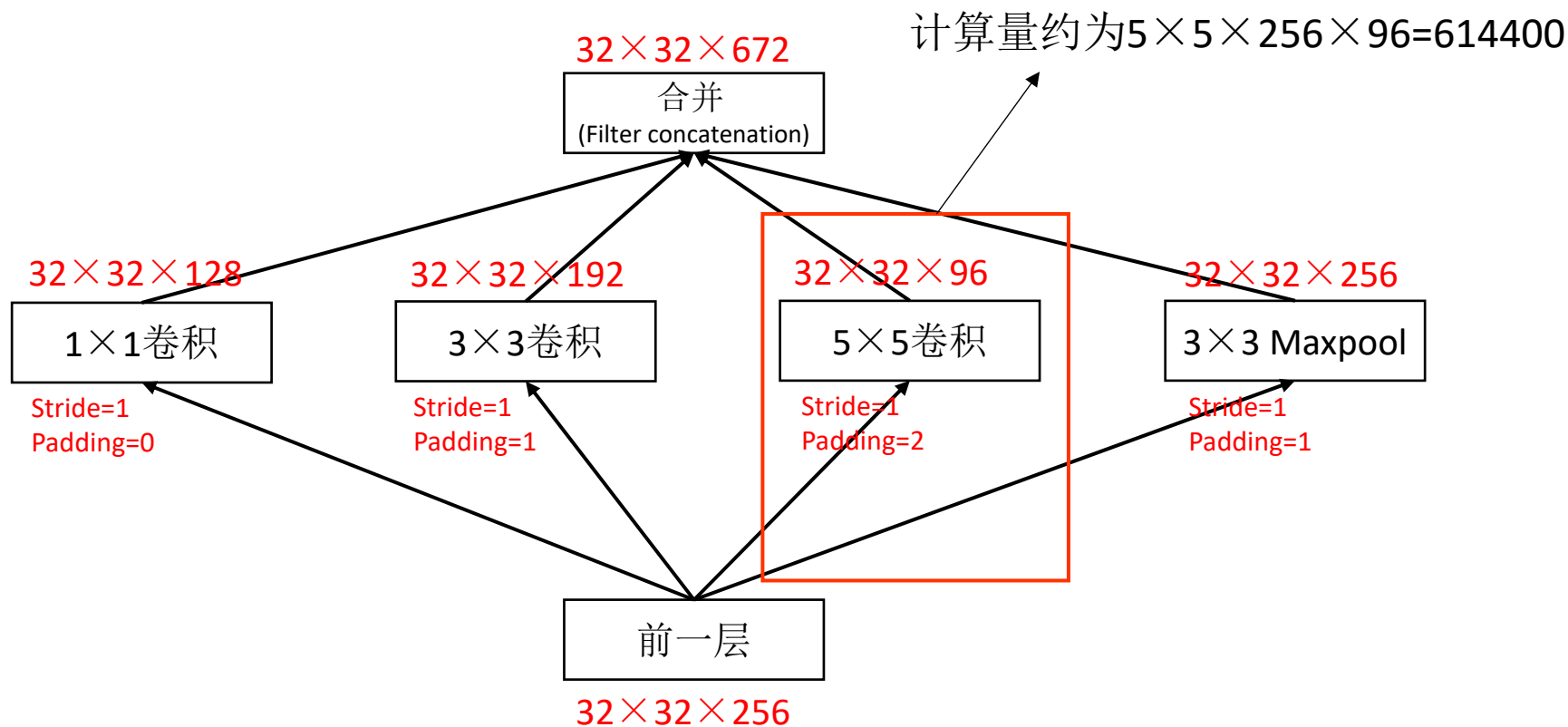
Inception 模块



■ GoogLeNet-- Inception模块

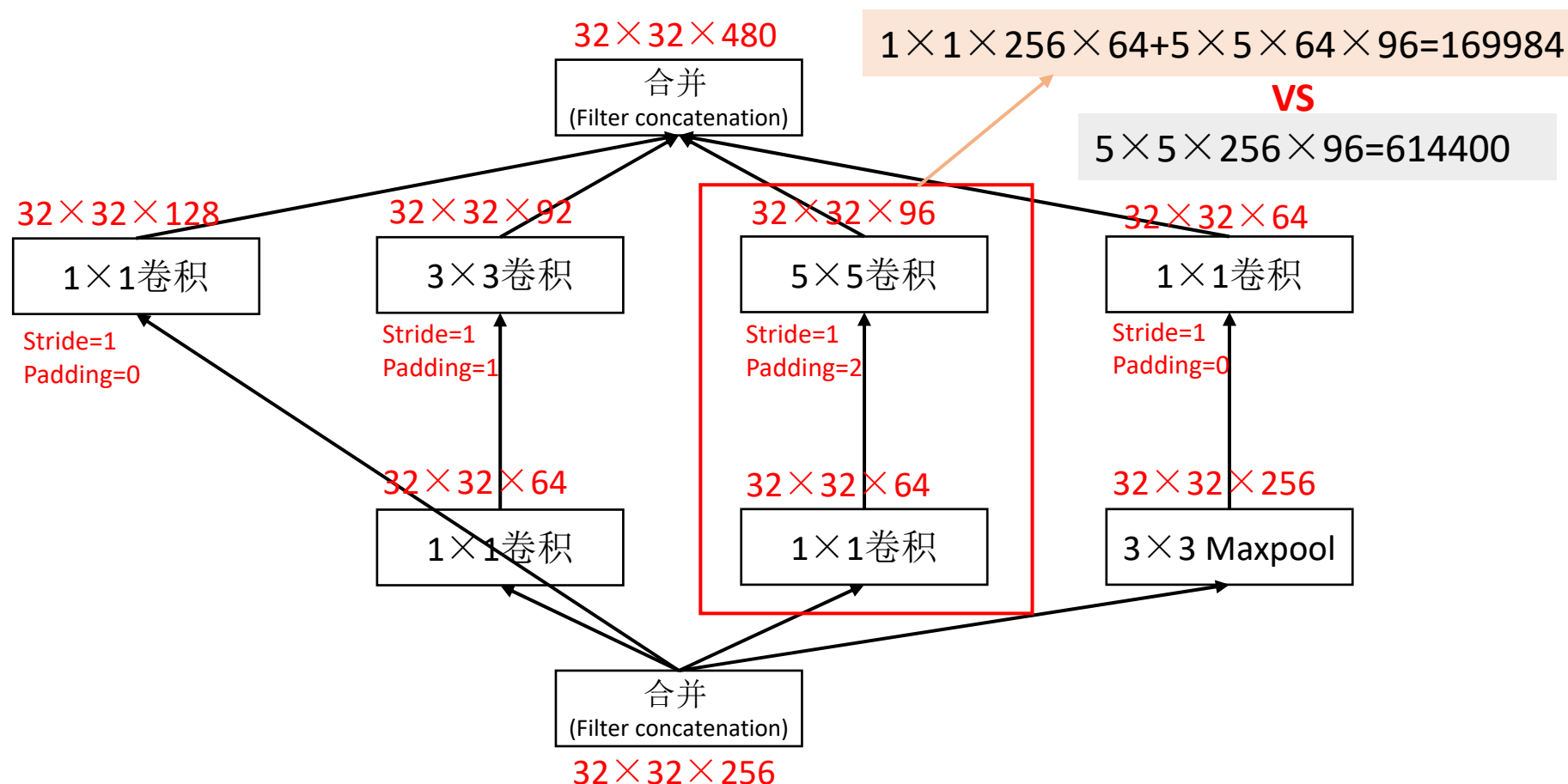
➤ Naïve version

- 5×5 卷积计算量太大



■ GoogLeNet-- Inception模块

- 解决：采用 1×1 卷积进行降维 (Stride=1, Padding=0)



■ GoogLeNet

➤ 网络参数表

type	patch size/ stride	output size	depth	#1×1	#3×3 reduce	#3×3	#5×5 reduce	#5×5	pool proj	params	ops
convolution	7×7/2	112×112×64	1							2.7K	34M
max pool	3×3/2	56×56×64	0								
convolution	3×3/1	56×56×192	2		64	192				112K	360M
max pool	3×3/2	28×28×192	0								
inception (3a)		28×28×256	2	64	96	128	16	32	32	159K	128M
inception (3b)		28×28×480	2	128	128	192	32	96	64	380K	304M
max pool	3×3/2	14×14×480	0								
inception (4a)		14×14×512	2	192	96	208	16	48	64	364K	73M
inception (4b)		14×14×512	2	160	112	224	24	64	64	437K	88M
inception (4c)		14×14×512	2	128	128	256	24	64	64	463K	100M
inception (4d)		14×14×528	2	112	144	288	32	64	64	580K	119M
inception (4e)		14×14×832	2	256	160	320	32	128	128	840K	170M
max pool	3×3/2	7×7×832	0								
inception (5a)		7×7×832	2	256	160	320	32	128	128	1072K	54M
inception (5b)		7×7×1024	2	384	192	384	48	128	128	1388K	71M
avg pool	7×7/1	1×1×1024	0								
dropout (40%)		1×1×1024	0								
linear		1×1×1000	1							1000K	1M
softmax		1×1×1000	0								

Fun features:

- Only 5 million params!
(Removes FC layers completely)

Compared to AlexNet:

- 12X less params
- 2x more compute
- 6.67% (vs. 16.4%)



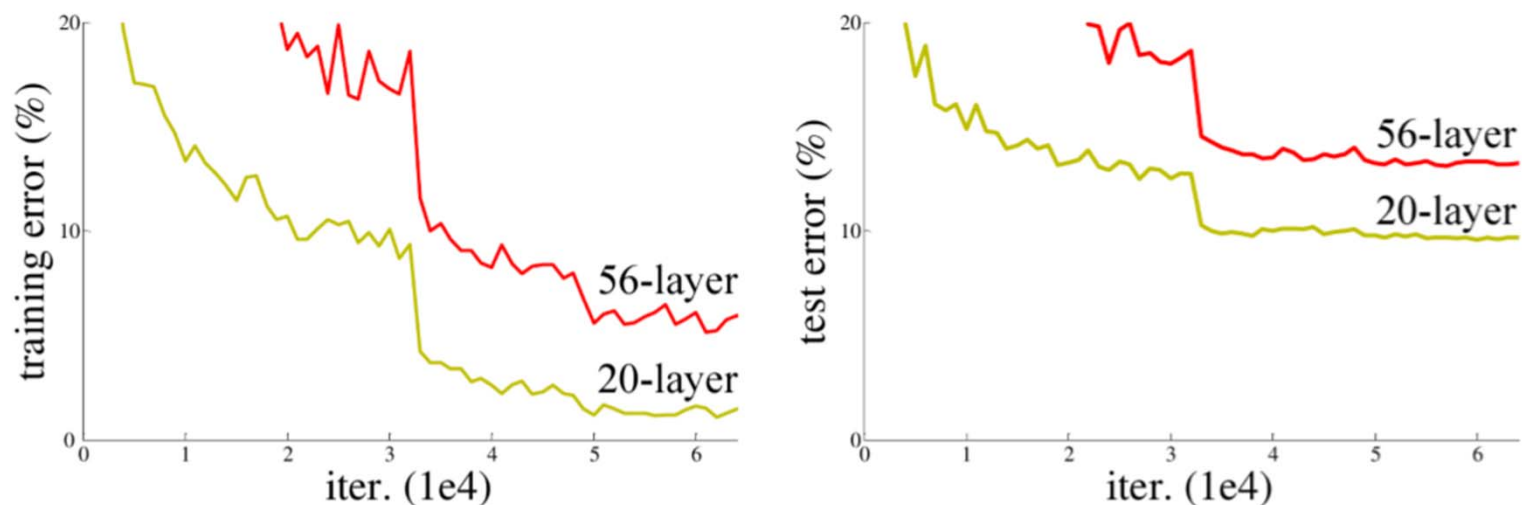
 大连理工大学
Dalian University Of Technology

- **残差网络 (Residual Network, ResNet)**
- **2015年ILSVRC比赛分类任务第一名 (152层, 错误率: 3.6%)**



■ ResNet

➤ 思考：随着网络层数的增加，网络性能为何下降？

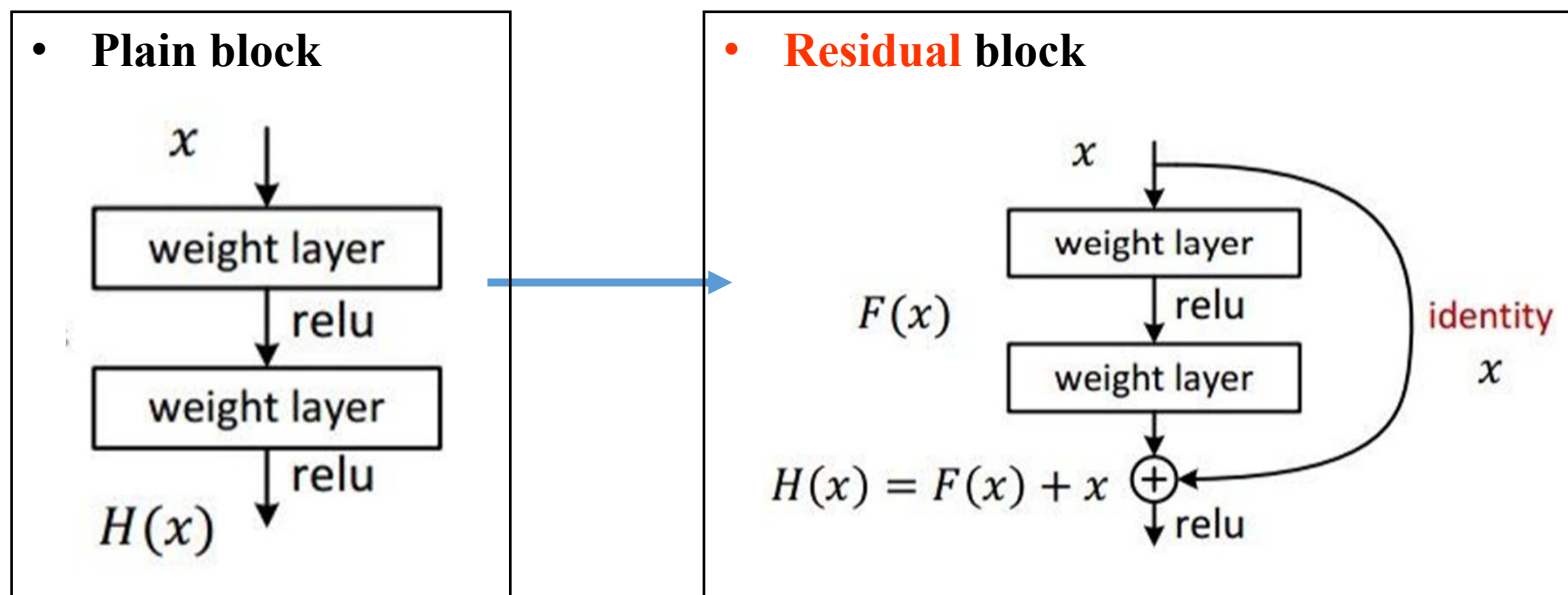


- 如果只是一味地对模型深度进行机械式累加，模型会出现梯度消失/爆炸、极易过拟合等模型性能退化的问题
- ResNet: 为解决网络深度增加后出现的性能退化问题



■ ResNet

- 残差单元：给卷积层增加跳连(Skip Connection)，以提升信息的传播效率，使网络更容易优化



■ ResNet

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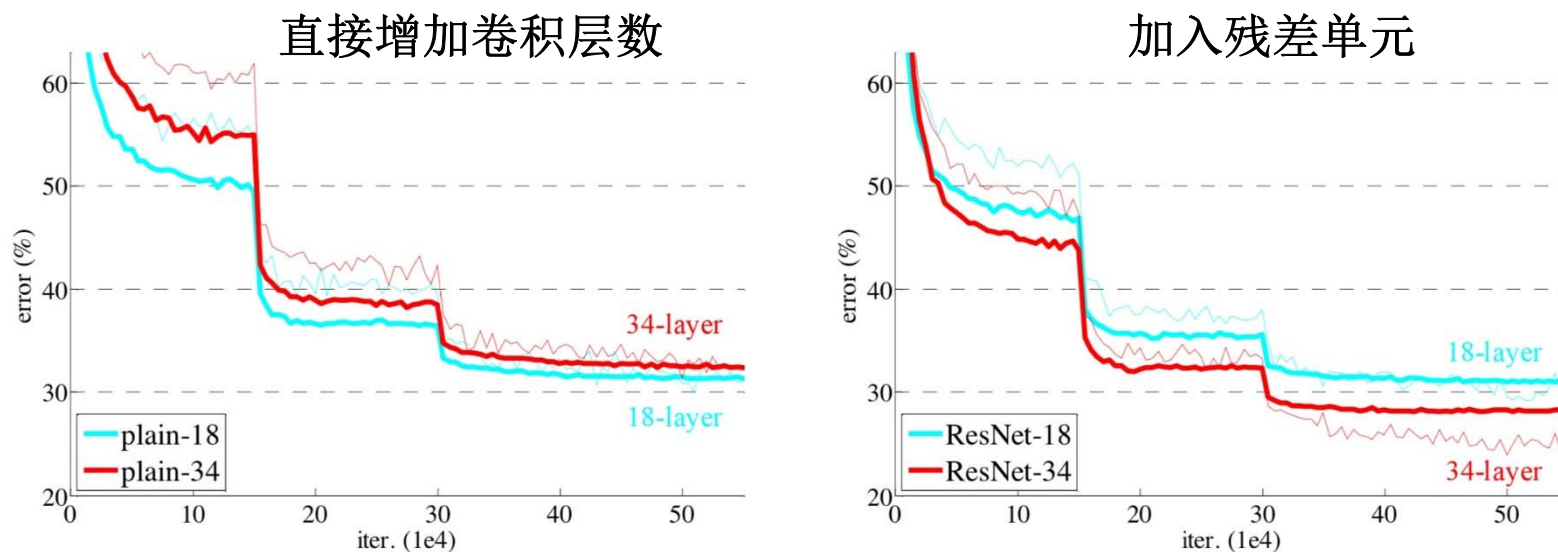
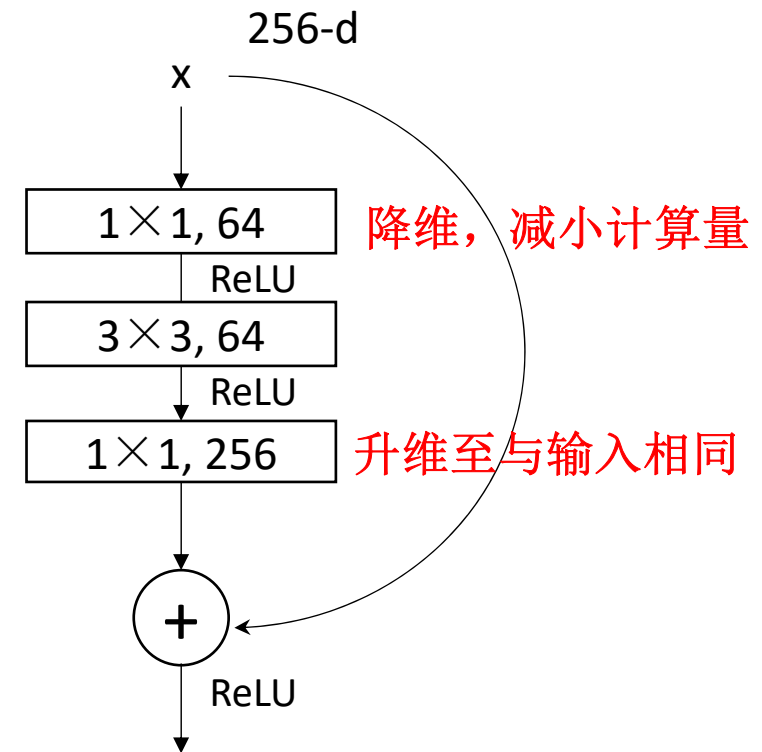
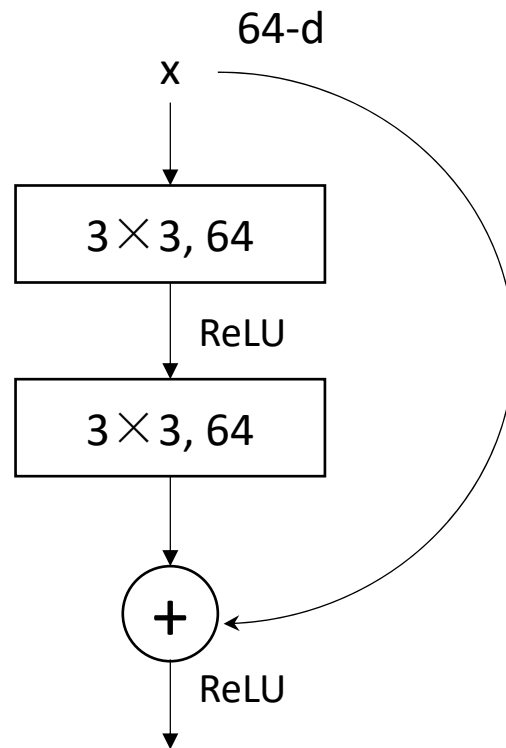


Figure 4. Training on **ImageNet**. Thin curves denote training error, and bold curves denote validation error of the center crops. Left: plain networks of 18 and 34 layers. Right: ResNets of 18 and 34 layers. In this plot, the residual networks have no extra parameter compared to their plain counterparts.



■ ResNet

- 残差单元：给卷积层增加**跳连(Skip Connection)**，以提升信息的传播效率，**使网络更容易优化**



基础网络 (Backbones)

■ ResNet

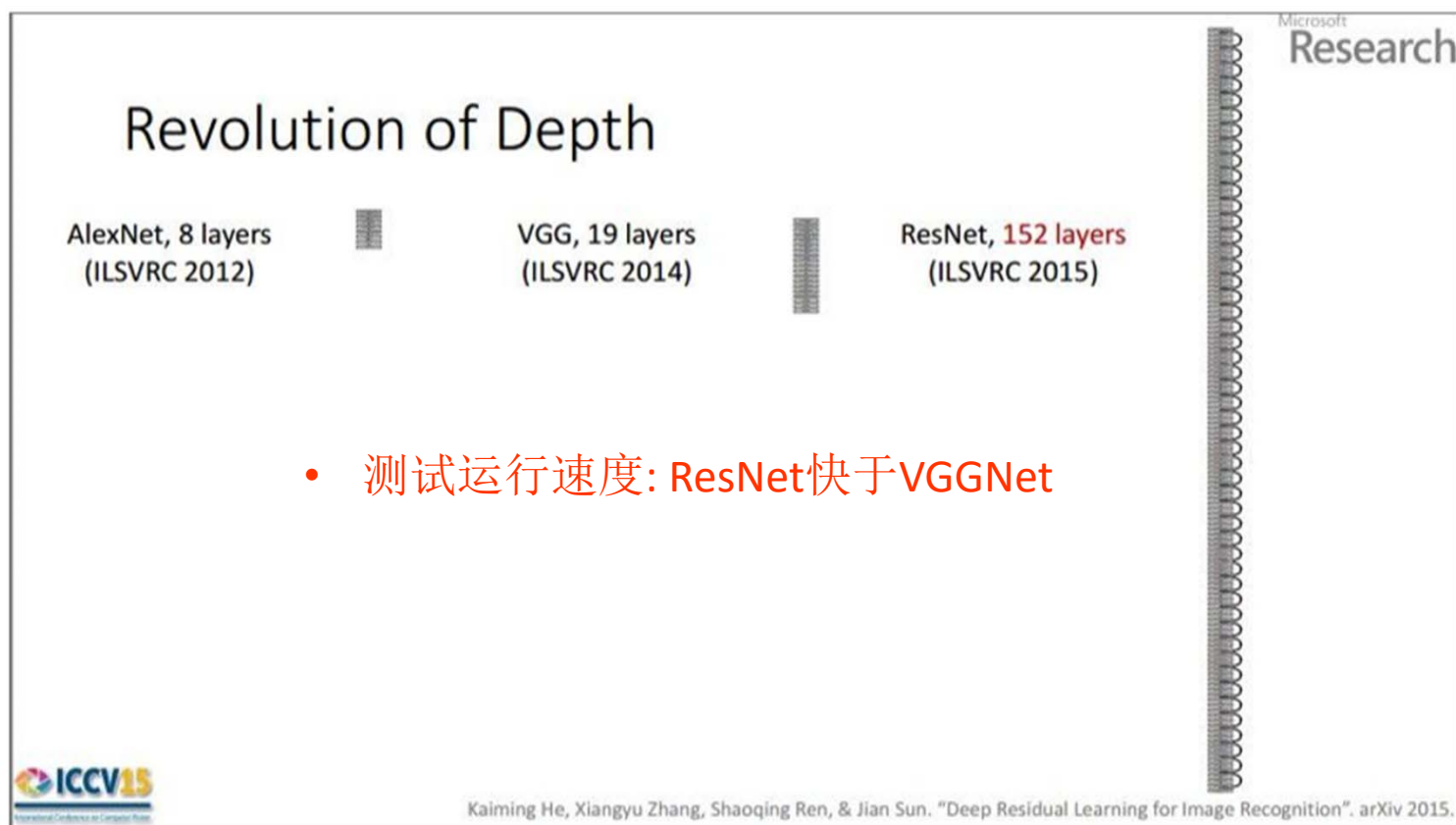
➤ 网络参数表

layer name	output size	18-layer	34-layer	50-layer	101-layer	152-layer
conv1	112×112	7×7, 64, stride 2				
conv2_x	56×56	3×3 max pool, stride 2				
		$\begin{bmatrix} 3 \times 3, 64 \\ 3 \times 3, 64 \end{bmatrix} \times 2$	$\begin{bmatrix} 3 \times 3, 64 \\ 3 \times 3, 64 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 64 \\ 3 \times 3, 64 \\ 1 \times 1, 256 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 64 \\ 3 \times 3, 64 \\ 1 \times 1, 256 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 64 \\ 3 \times 3, 64 \\ 1 \times 1, 256 \end{bmatrix} \times 3$
conv3_x	28×28	$\begin{bmatrix} 3 \times 3, 128 \\ 3 \times 3, 128 \end{bmatrix} \times 2$	$\begin{bmatrix} 3 \times 3, 128 \\ 3 \times 3, 128 \end{bmatrix} \times 4$	$\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 128 \\ 1 \times 1, 512 \end{bmatrix} \times 4$	$\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 128 \\ 1 \times 1, 512 \end{bmatrix} \times 4$	$\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 128 \\ 1 \times 1, 512 \end{bmatrix} \times 8$
conv4_x	14×14	$\begin{bmatrix} 3 \times 3, 256 \\ 3 \times 3, 256 \end{bmatrix} \times 2$	$\begin{bmatrix} 3 \times 3, 256 \\ 3 \times 3, 256 \end{bmatrix} \times 6$	$\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256 \\ 1 \times 1, 1024 \end{bmatrix} \times 6$	$\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256 \\ 1 \times 1, 1024 \end{bmatrix} \times 23$	$\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256 \\ 1 \times 1, 1024 \end{bmatrix} \times 36$
conv5_x	7×7	$\begin{bmatrix} 3 \times 3, 512 \\ 3 \times 3, 512 \end{bmatrix} \times 2$	$\begin{bmatrix} 3 \times 3, 512 \\ 3 \times 3, 512 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 512 \\ 3 \times 3, 512 \\ 1 \times 1, 2048 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 512 \\ 3 \times 3, 512 \\ 1 \times 1, 2048 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 512 \\ 3 \times 3, 512 \\ 1 \times 1, 2048 \end{bmatrix} \times 3$
	1×1	average pool, 1000-d fc, softmax				
FLOPs		1.8×10^9	3.6×10^9	3.8×10^9	7.6×10^9	11.3×10^9



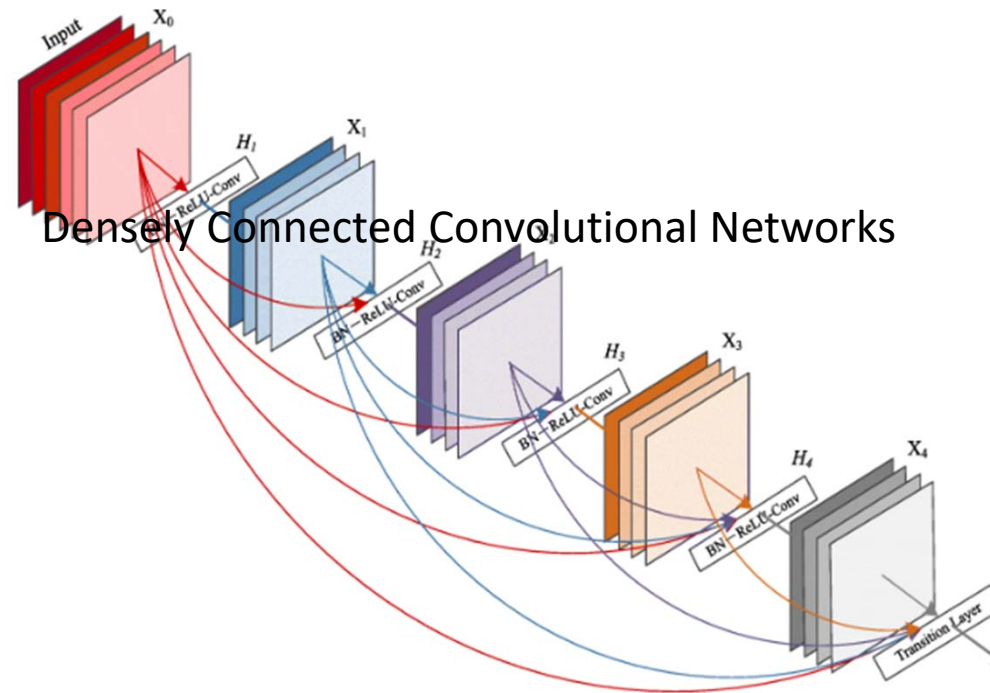
■ ResNet

ILSVRC 2015 winner (3.6% top 5 error)



■ DenseNet

- 前面所有层与后面层均密集连接 (Dense Connection)

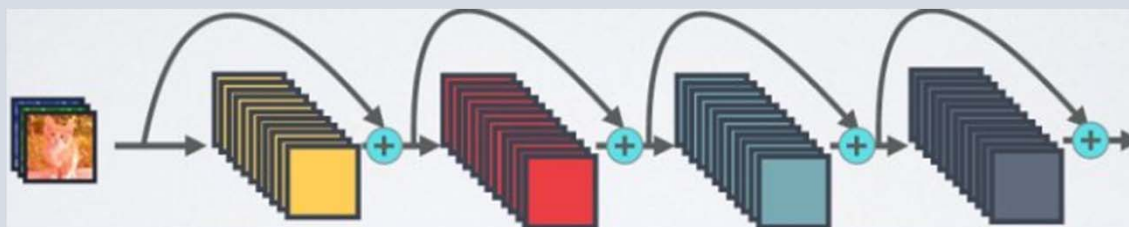


Huang *et al.*, Densely Connected Convolutional Networks, CVPR2017.

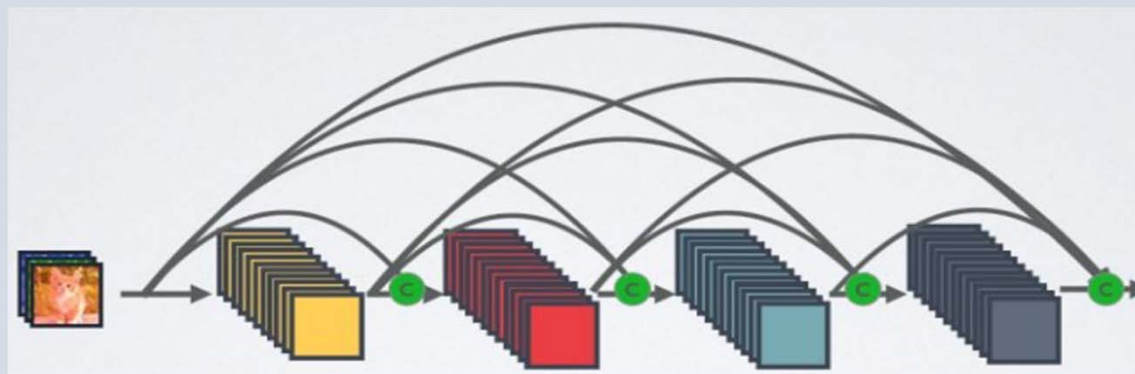


■ DenseNet

- 前面所有层与后面层均密集连接 (**Dense Connection**)
- 通过特征在**channel**上的连接来实现特征重用 (**Feature Reuse**)



ResNet的短路连接机制，“+”表示元素级相加操作(Element-wise addition)



DenseNet的密集连接机制，“c”表示通道维度的合并操作(Channel-wise concatenation)



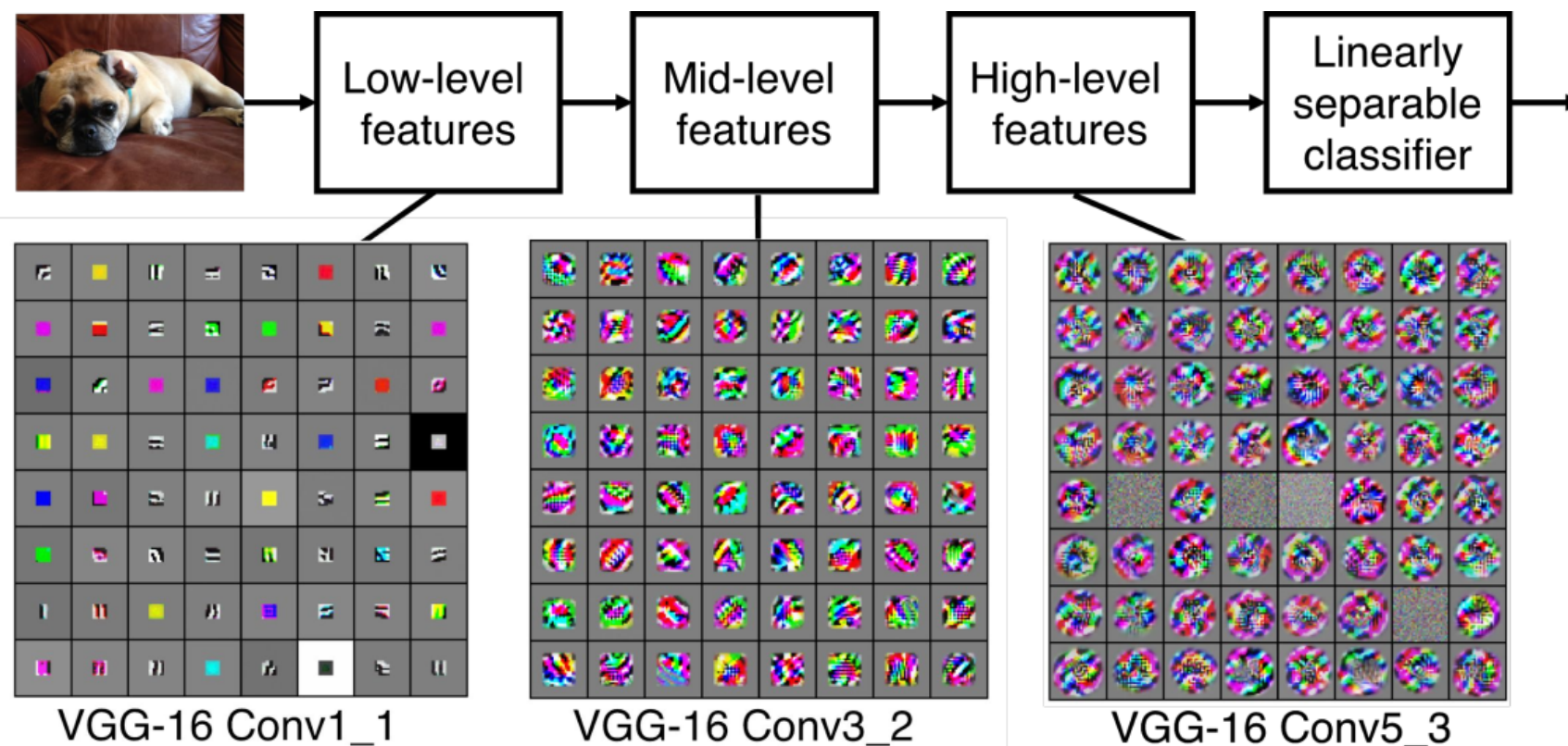
PART TWO

卷积网络可视化 CNN Visualization



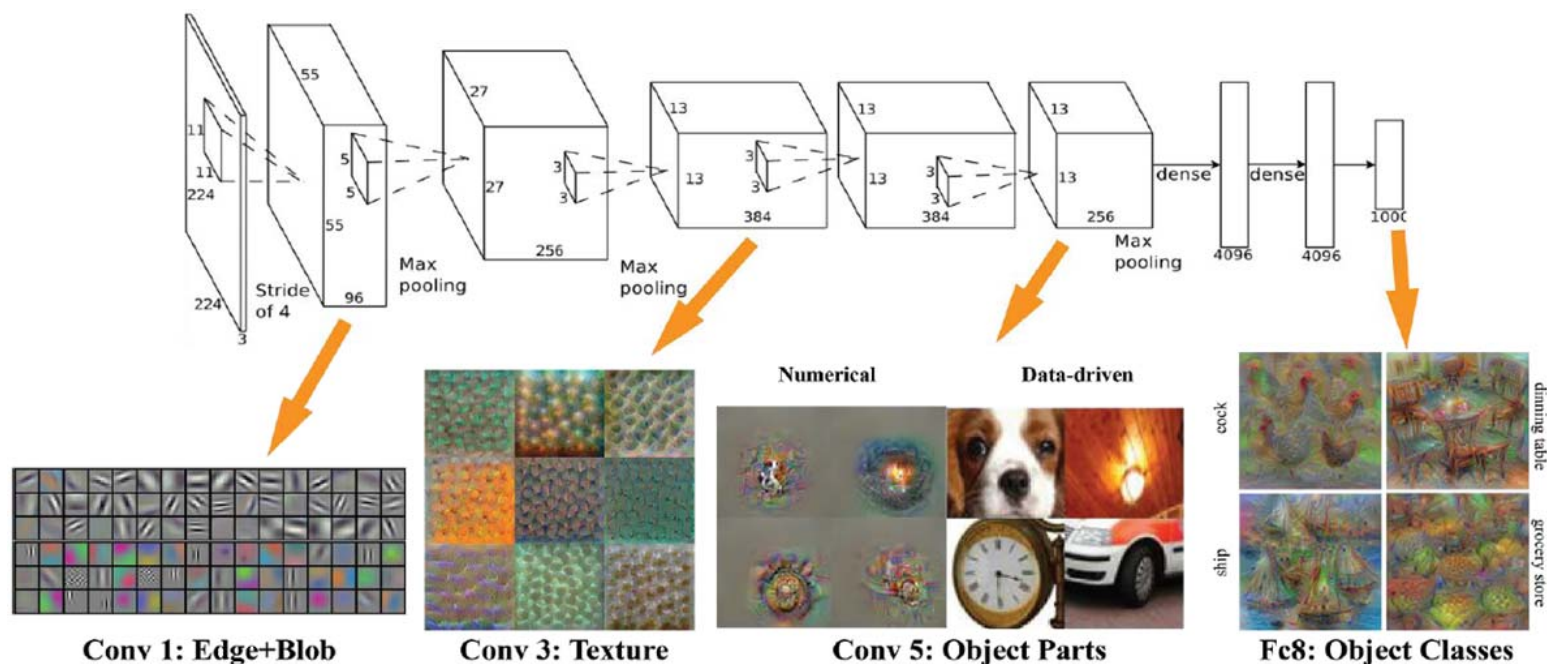
可视化 (Visualization)

- 不同层级的滤波器提取到的feature maps不同



可视化 (Visualization)

- 不同层级的卷积层所学习到的图像特征具有**分层的特性**
 - 浅层的卷积层学习到图像的**低级 (Low-level)** 特征，如颜色、边缘、纹理
 - 深层的卷积层学到图像的高级 (**High-level**) 特征，如物体位置、语义类别



用于图像分类的卷积网络



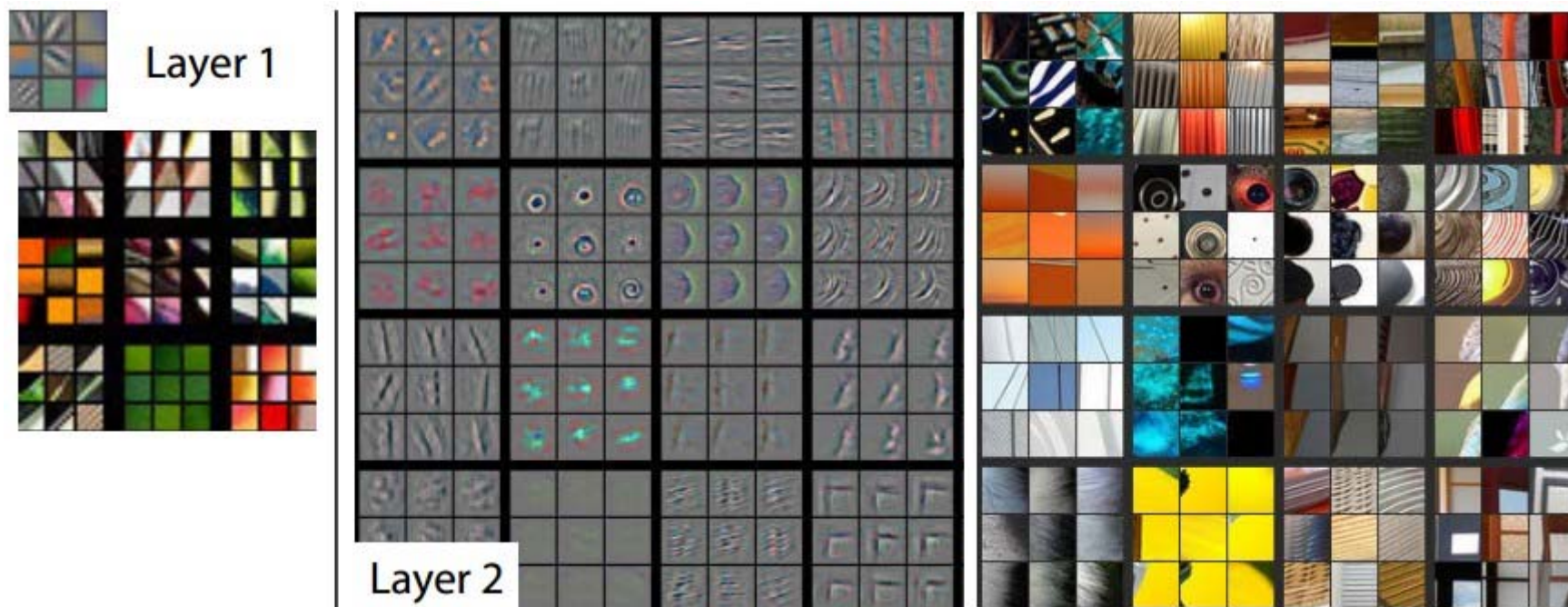
可视化 (Visualization)

- AlexNet中第一层卷积的滤波器 (96 filters [11x11x3])



可视化 (Visualization)

- 第二层响应了角落和其他的边缘/颜色信息

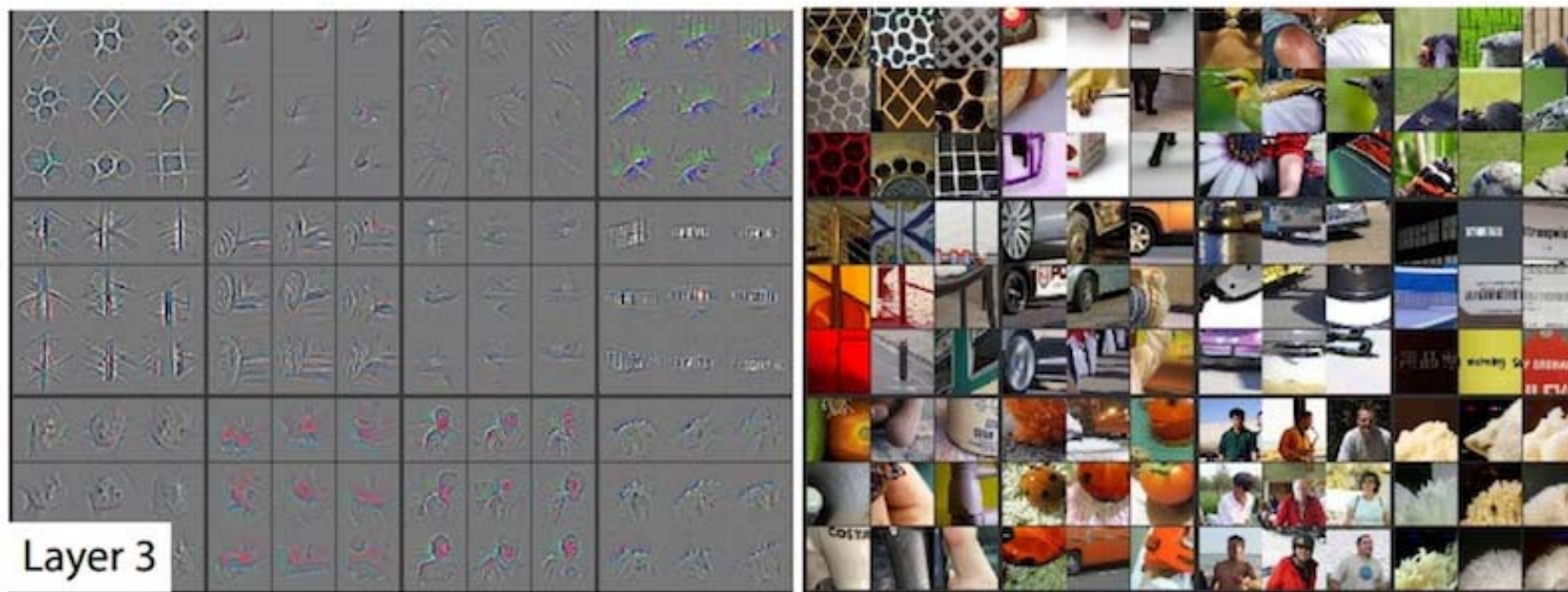


Zeiler *et al.*, Visualizing and Understanding Convolutional Networks, ECCV2014.



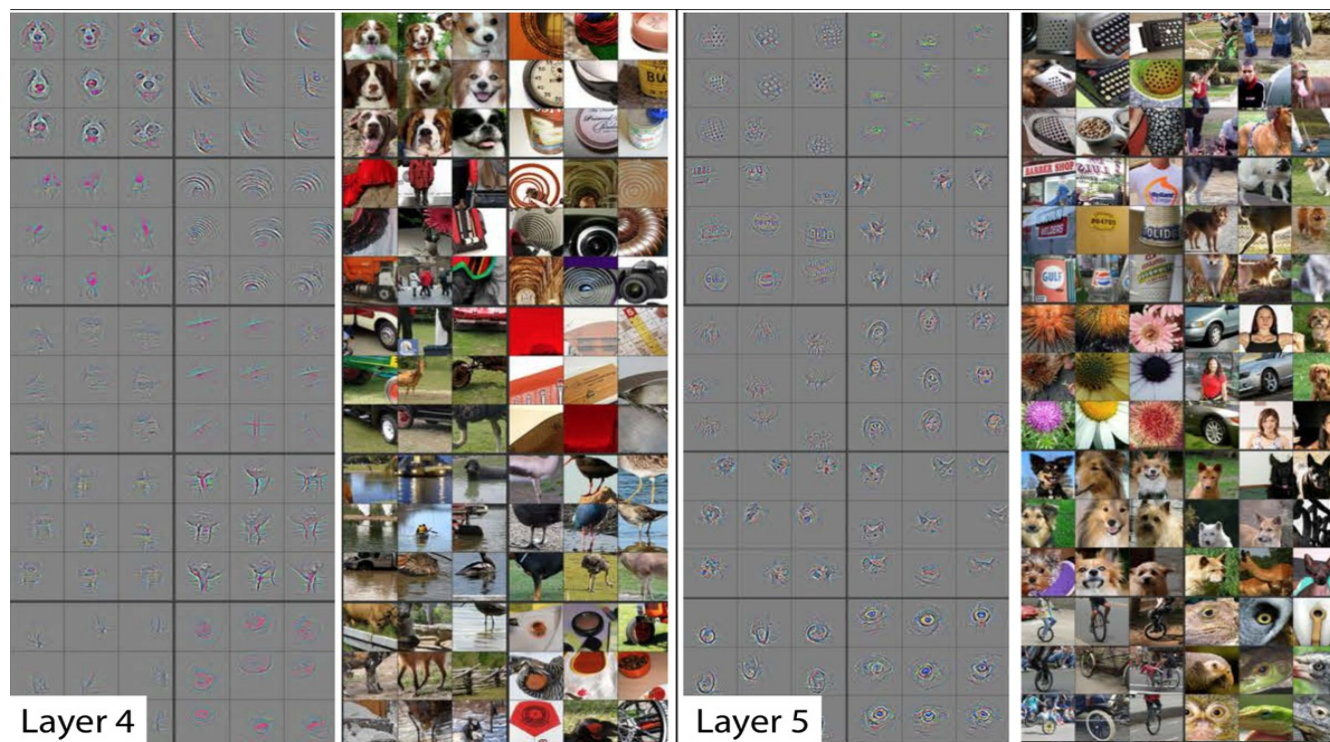
可视化 (Visualization)

- 第三层具有更复杂的不变性，捕获相似的纹理



可视化 (Visualization)

- 第四层显示了显著的变化，并且更加类别具体化
- 第五层显示了具有显著姿态变化的整个对象



PART THREE

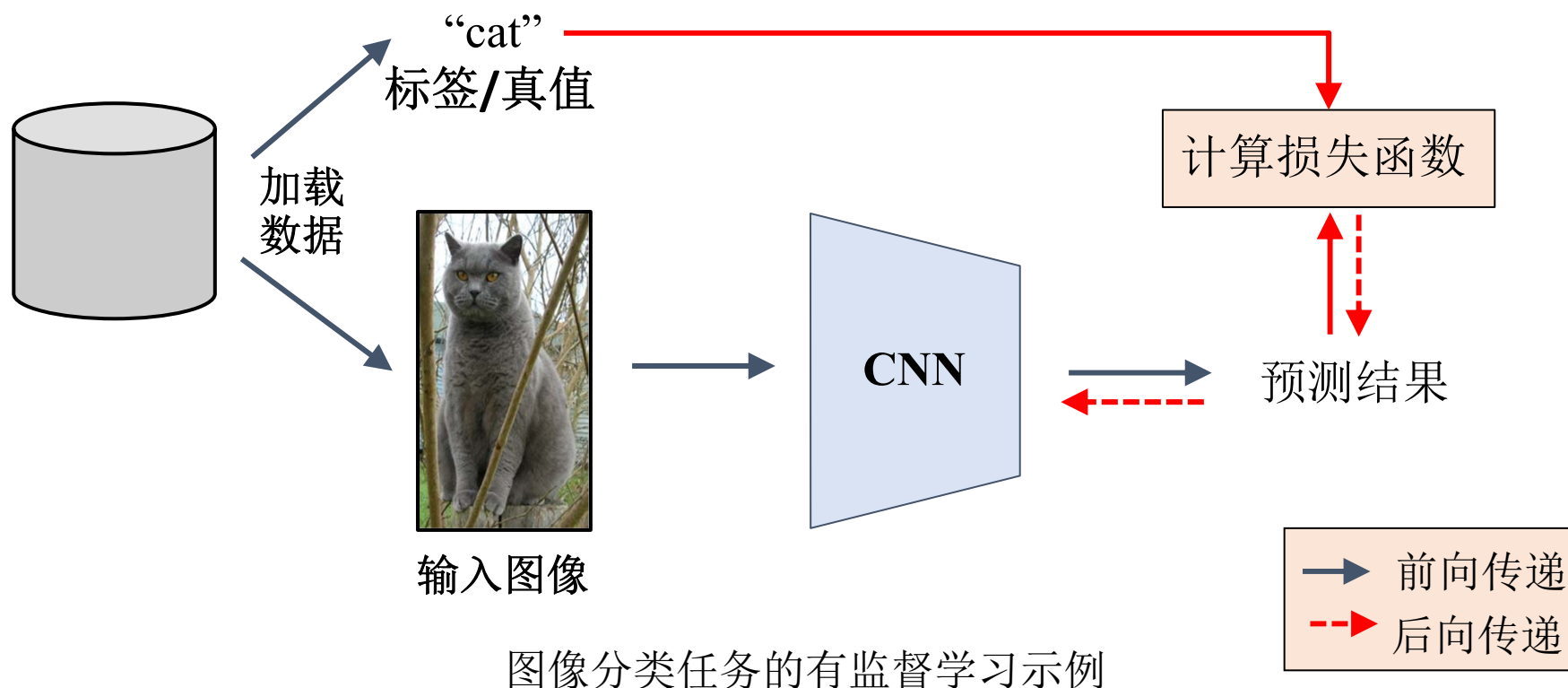
预训练与微调

Pre-training and Fine-tuning



预训练与微调 (Pre-training & Fine-tuning)

➤ CNN有监督学习



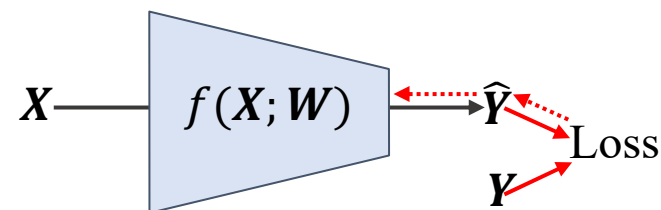
- 当训练数据量少时，容易出现过拟合、优化陷入局部极小值、模型性能差、泛化能力低等风险。
- 解决方案
 - 采用数据扩增 (Data Augmentation)，增加数据多样性
 - 预训练(Pre-training) + 微调(Fine-tuning)策略



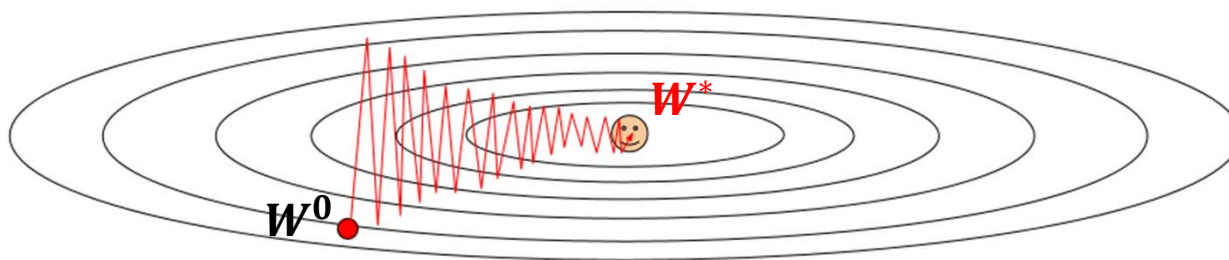
预训练与微调 (Pre-training & Fine-tuning)

➤ 举例说明

- 给定目标任务，搭建的卷积网络为： $f(X; W)$



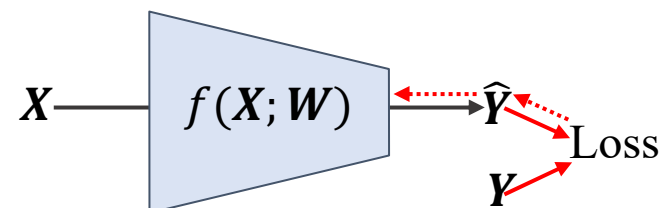
- 1) 首先对 W 进行随机初始化，如服从分布 $\mathcal{N}(0,1)$ 。假设 W 的初始化值为 W^0
- 2) 通过最小化损失函数，进行后向传递，不断对 W 更新直至找到最优解 W^*



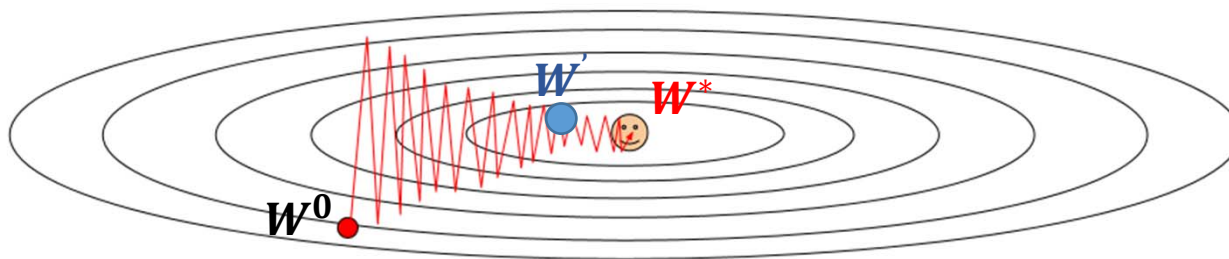
预训练与微调 (Pre-training & Fine-tuning)

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- 给定目标任务，搭建的卷积网络为： $f(X; W)$

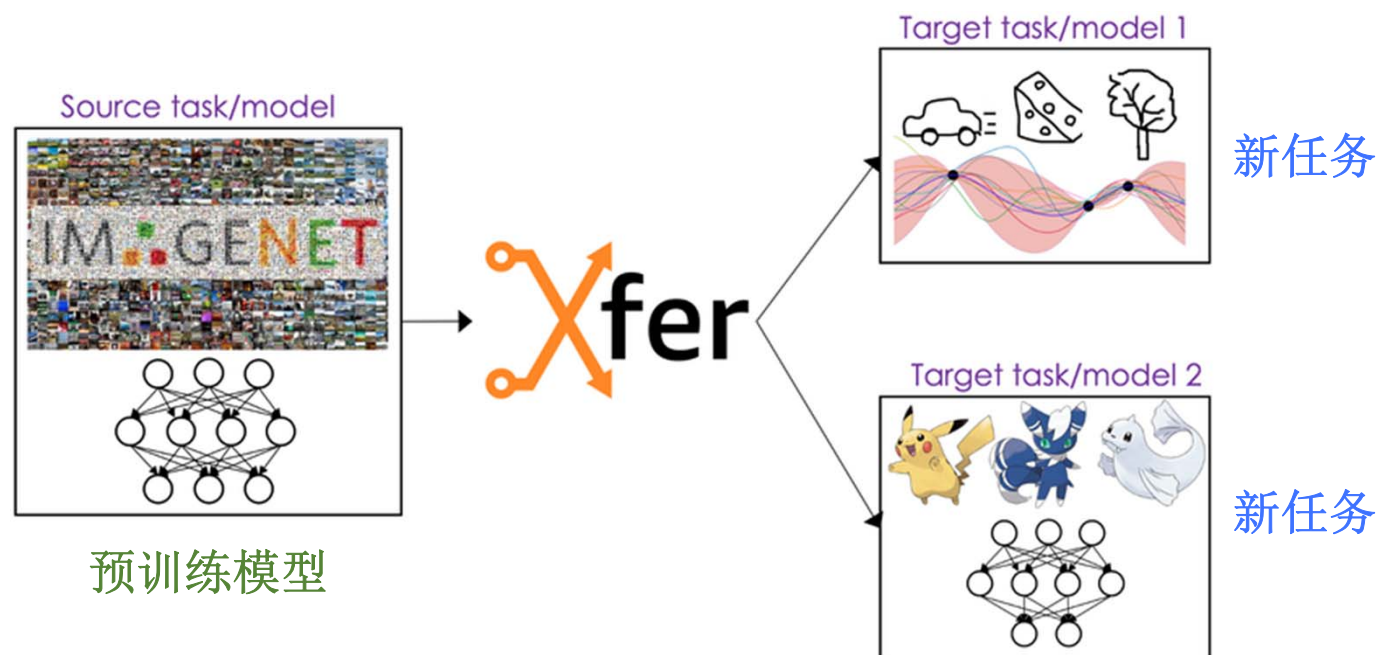


- 1) 首先对 W 进行随机初始化，如服从分布 $\mathcal{N}(0,1)$ 。假设 W 的初始化值为 W^0
 - 2) 通过最小化损失函数，进行后向传递，不断对 W 更新直至找到最优解 W^*
 - 3) 若在步骤 1开始前，为你提供靠近 W^* 的参数 W'
 - 4) 将 W' 作为 W 的初始化值，并优化学习 (步骤2)，此时网络训练时间将大大缩短
- 步骤3相当于提供一个预训练模型 (Pre-trained model)
 - 步骤4则为基于该模型的微调 (Fine-tuning)



预训练与微调 (Pre-training & Fine-tuning)

- 使用已在大数据集上经过训练的模型（预训练模型），可直接使用相应的结构和权重，将它们应用到新任务上
- 好处：节省新任务的训练时间、提高收敛速度、提高结果精度



预训练与微调 (Pre-training & Fine-tuning)



- 图像分类为其他任务提供了诸多优秀的预训练模型（如VGG、ResNet等）

Model	Size	Top-1 Accuracy	Top-5 Accuracy	Parameters	Depth
Xception	88 MB	0.790	0.945	22,910,480	126
VGG16	528 MB	0.715	0.901	138,357,544	23
VGG19	549 MB	0.727	0.910	143,667,240	26
ResNet50	99 MB	0.759	0.929	25,636,712	168
InceptionV3	92 MB	0.788	0.944	23,851,784	159
InceptionResNetV2	215 MB	0.804	0.953	55,873,736	572
MobileNet	17 MB	0.665	0.871	4,253,864	88
DenseNet121	33 MB	0.745	0.918	8,062,504	121
DenseNet169	57 MB	0.759	0.928	14,307,880	169
DenseNet201	80 MB	0.770	0.933	20,242,984	201



预训练与微调 (Pre-training & Fine-tuning)



- 把预训练模型参数加载后，在新的任务上对这些模型参数进一步微调
- 根据新任务的数据量大小、与预训练数据集的差异性，可选择对新任务模型中的哪些网络层进行冻结(Freeze)、微调 (Fine-tune) 或 重新训练 (Retrain/Train from scratch)



预训练与微调 (Pre-training & Fine-tuning)

1. Trained on Imagenet



预训练与微调 (Pre-training & Fine-tuning)

1. Trained on Imagenet



2. Small dataset



预训练与微调 (Pre-training & Fine-tuning)

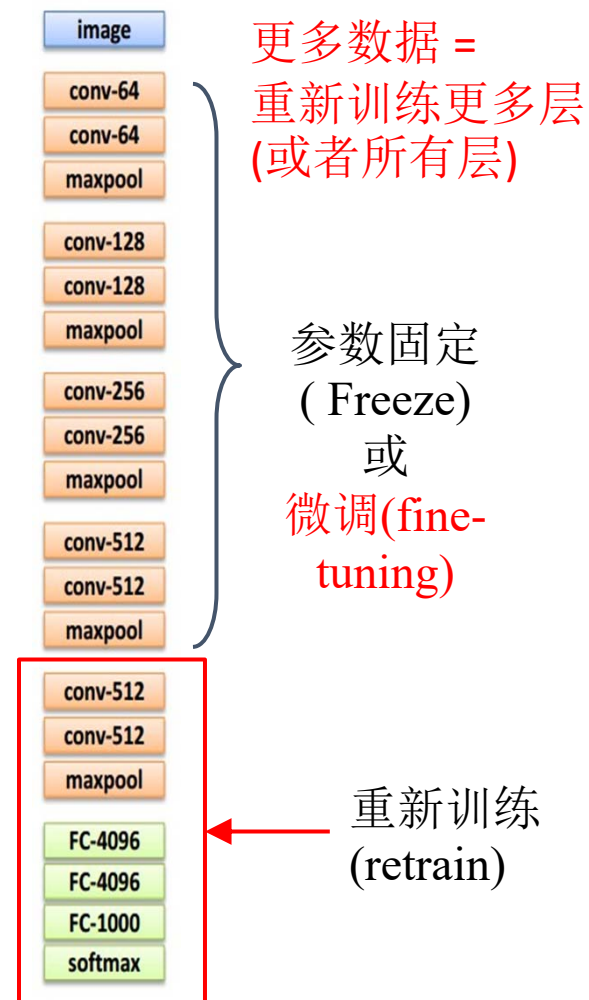
1. Trained on Imagenet



2. Small dataset



3. Bigger dataset

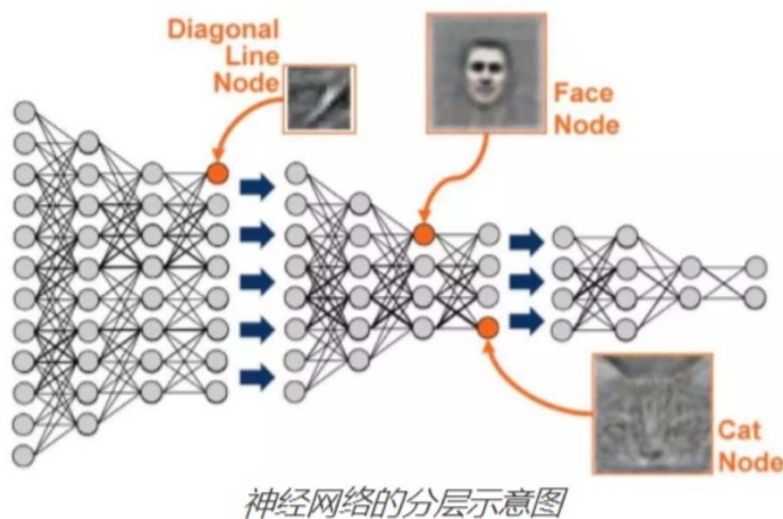


预训练与微调 (Pre-training & Fine-tuning)

➤ 为什么要微调？

卷积神经网络的核心：

- 浅层卷积层提取**基础特征**，如边缘、轮廓
- 深层卷积层提取**抽象特征**（如整个脸型）
- 全连接层根据**特征组合**进行评分分类



在不同任务间的泛化性

好

浅层

image

conv-64

conv-64

maxpool

conv-128

conv-128

maxpool

conv-256

conv-256

maxpool

conv-512

conv-512

maxpool

conv-512

conv-512

maxpool

FC-4096

FC-4096

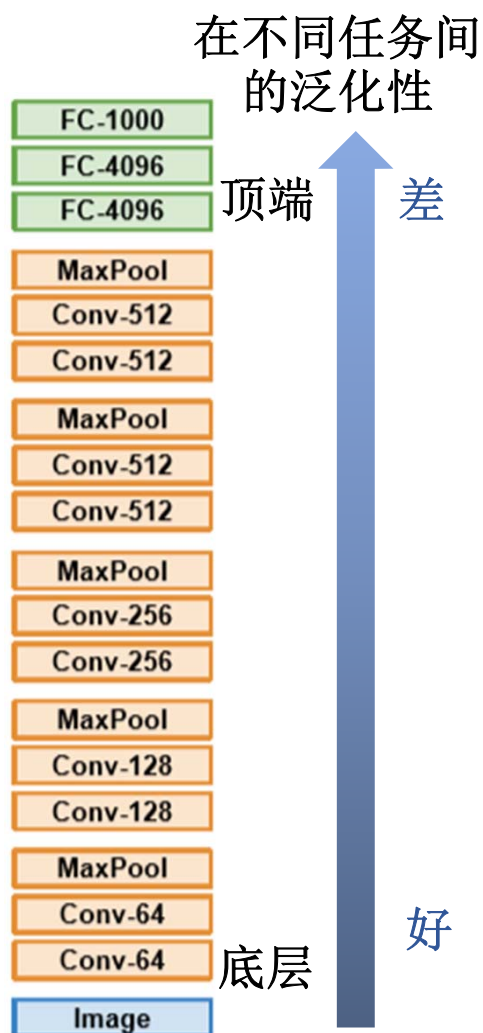
FC-1000

softmax

深层

差

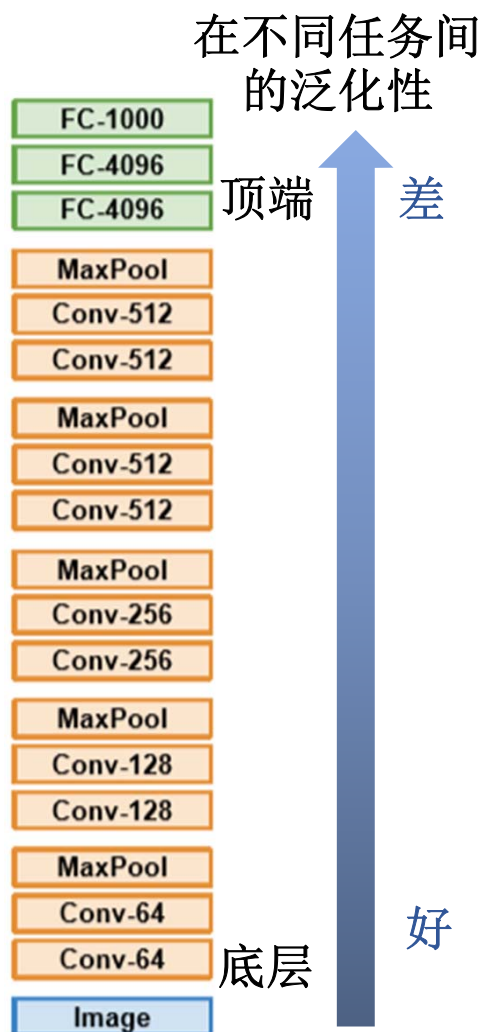
预训练与微调 (Pre-training & Fine-tuning)



新任务数据	与预训练数据 相似度高	与预训练数据 差异明显
少量	?	?
大量	?	?



预训练与微调 (Pre-training & Fine-tuning)



新任务数据	与预训练数据 相似度高	与预训练数据 差异明显
少量	仅微调顶端的线性分类层	You're in trouble... 采取数据扩增/ 收集更多的数据
大量	微调较多的卷积层	微调更多的卷积层/整个网络/从头训练 (Training from scratch)

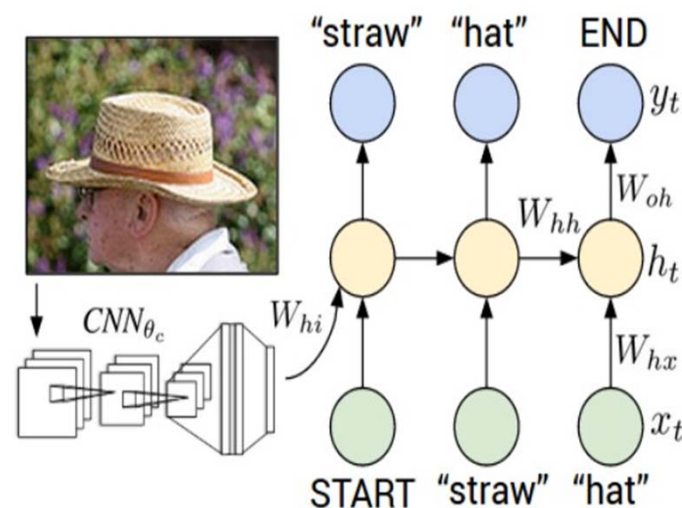
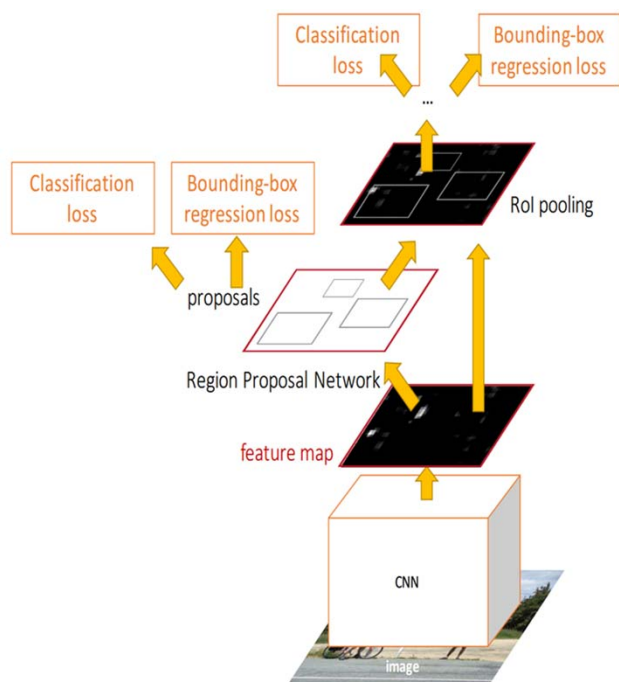
Tip: 微调时, 可采用比training from scratch 更小的学习率



预训练与微调 (Pre-training & Fine-tuning)

➤ CNN的预训练与微调策略应用非常普遍

- 目标检测(Object detection): Faster R-CNN
- 图像描述(Image captioning): CNN+RNN



Ren *et al.*, "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks", NIPS 2015.

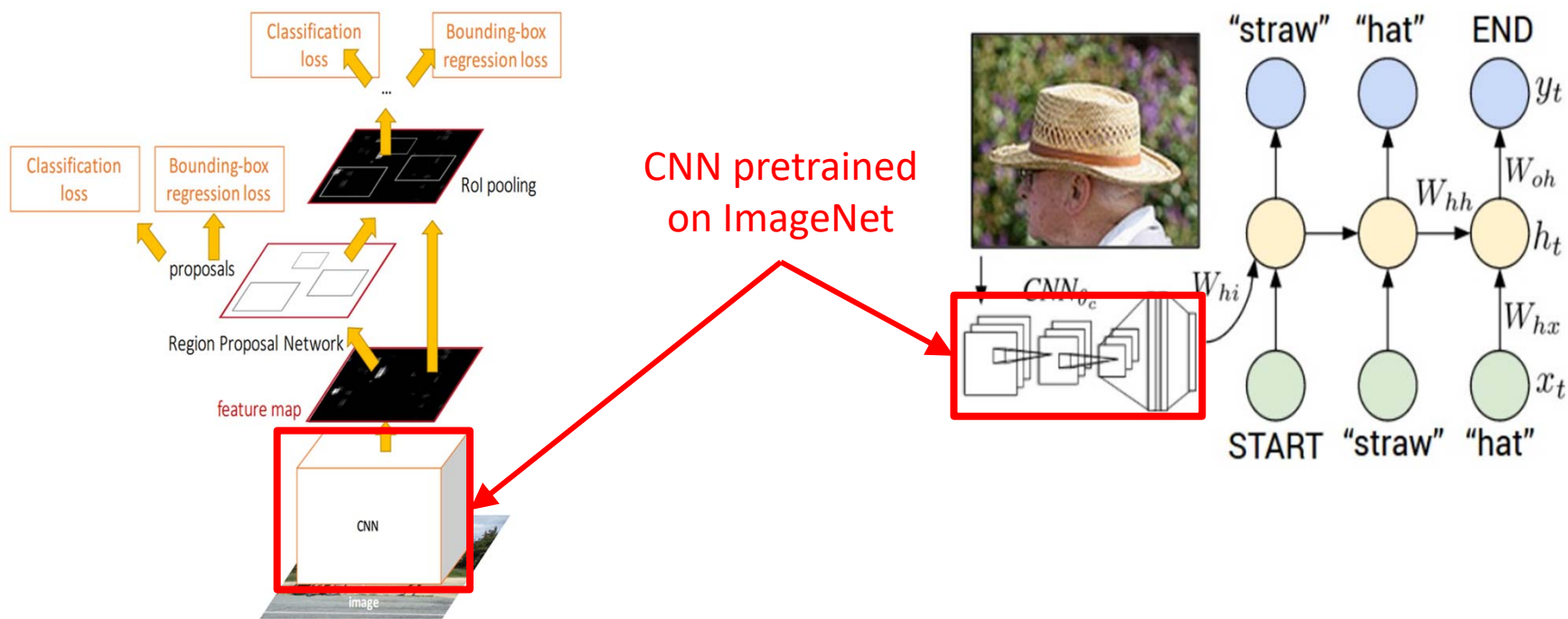
Karpathy *et al.*, "Deep Visual-Semantic Alignments for Generating Image Descriptions", CVPR2015.



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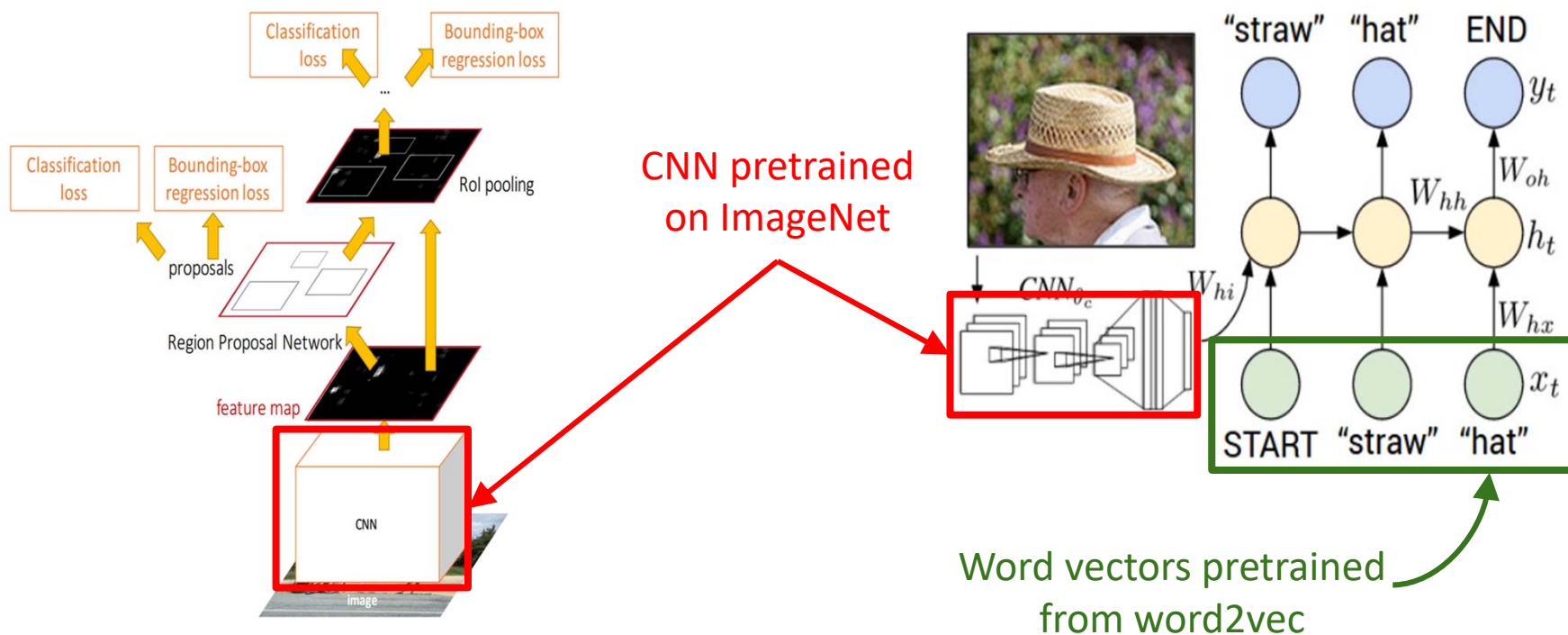
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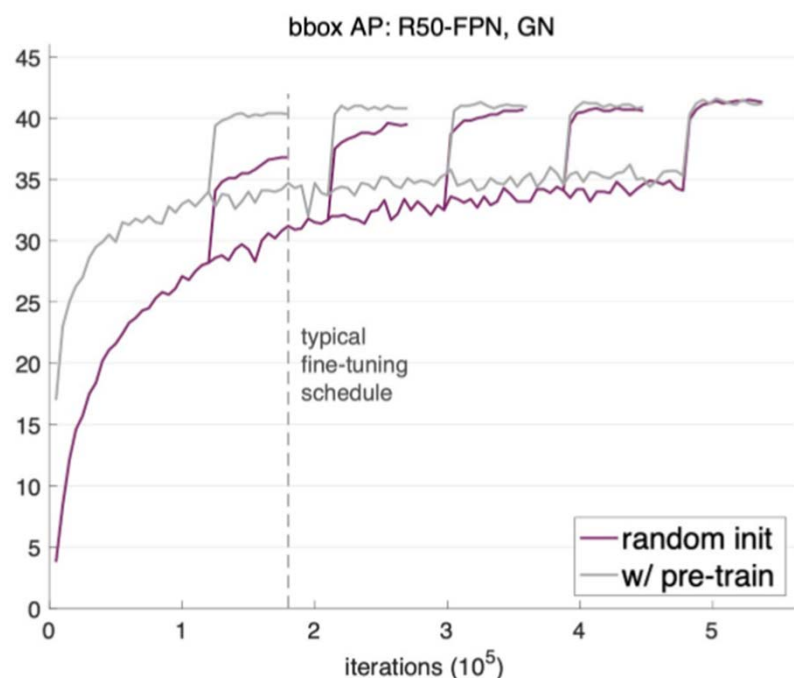
Karpathy *et al.*, "Deep Visual-Semantic Alignments for Generating Image Descriptions", CVPR2015.



预训练与微调 (Pre-training & Fine-tuning)

➤ CNN的预训练与微调策略应用非常普遍

然而，最新工作表明预训练**并非必要**的！



目标检测网络训练过程

- a. 网络参数随机初始化
- b. 网络参数用一个在ImageNet训练后的模型初始化
- 经过充足的训练，方式a能达到与方式b相当的性能，但前者的训练时间为后者的2~3倍
- 更大的训练数据比从一个相关任务模型上微调更有效

He et al, "Rethinking ImageNet Pre-training", ICCV 2019.



预训练与微调 (Pre-training & Fine-tuning)



➤ CNN的预训练与微调策略应用非常普遍

现有的深度学习框架提供了“**Model Zoo**”，里面含有很多训练好的模型

- TensorFlow: <https://github.com/tensorflow/models>
- PyTorch: <https://github.com/pytorch/vision>



04 PART FOUR

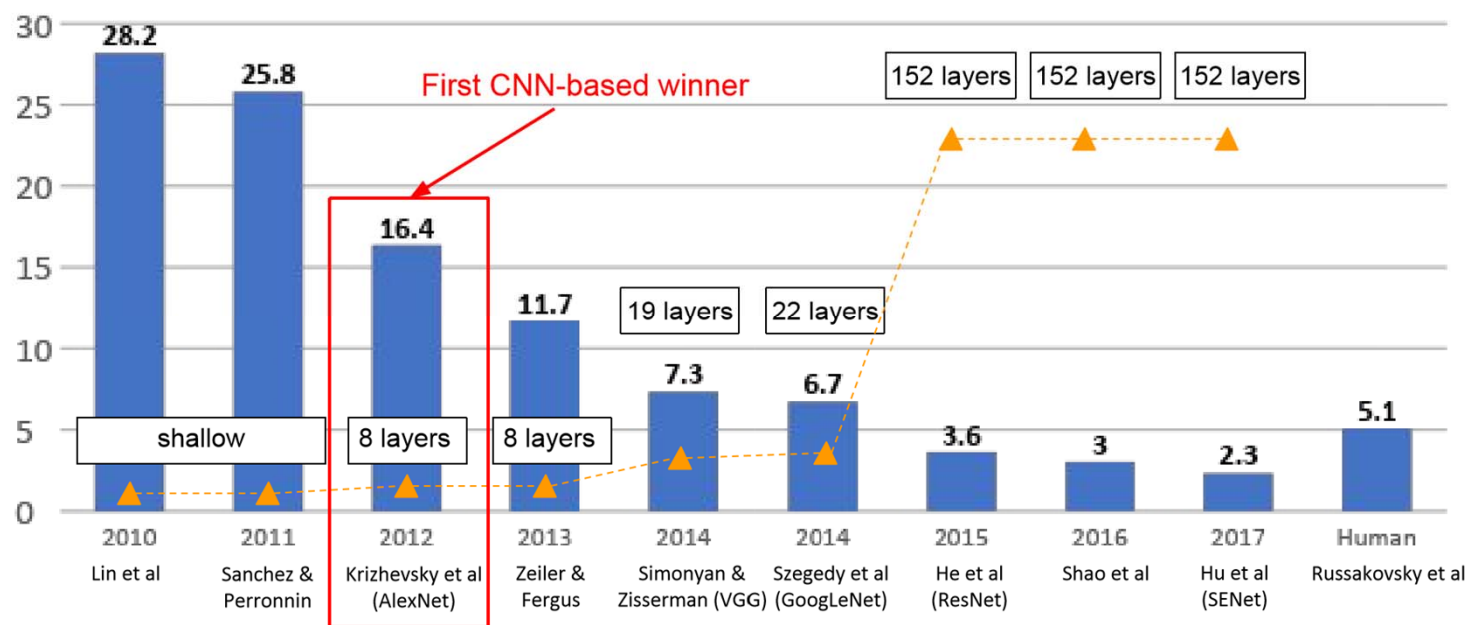
小结 Summary



小结 (Summary)

■ 基础网络

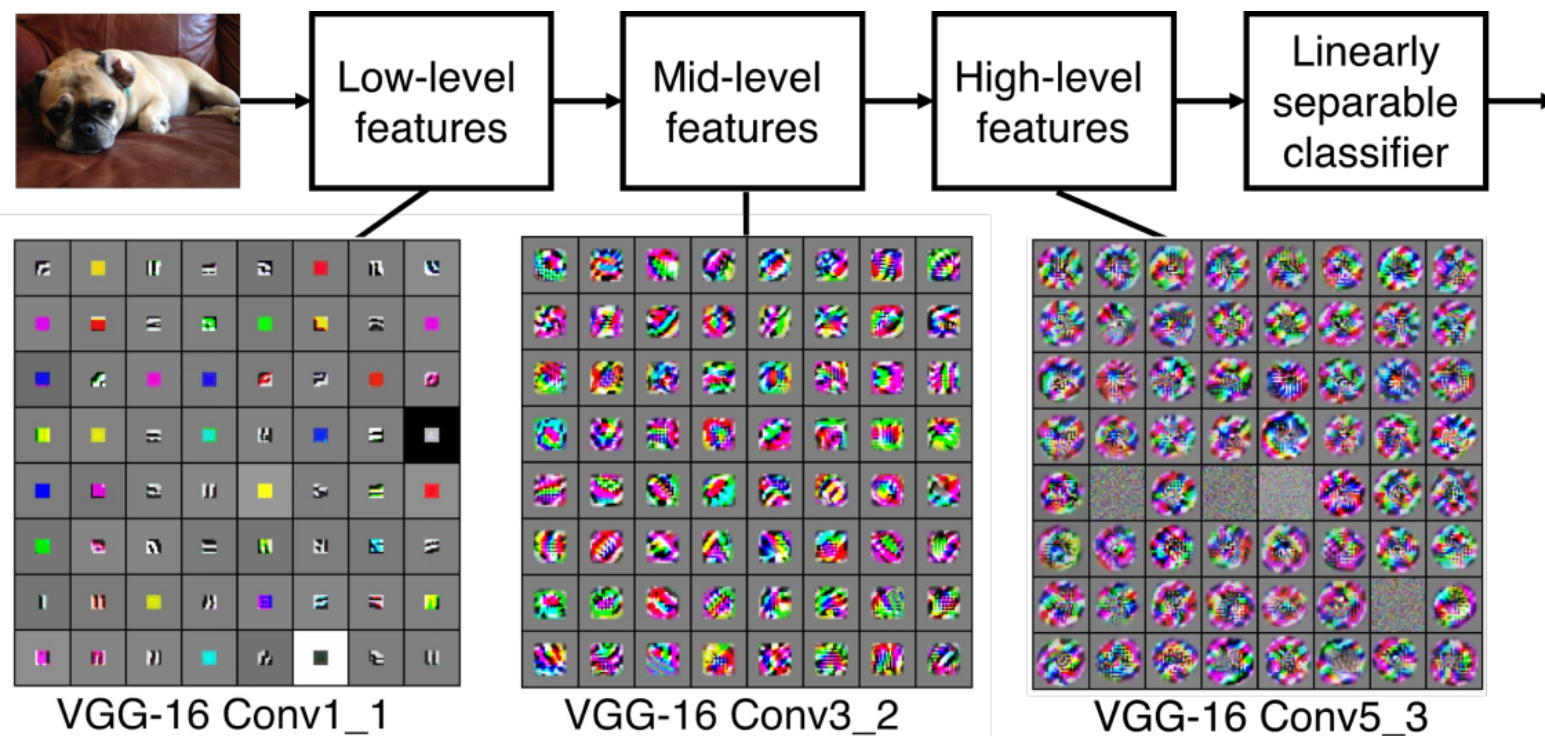
- AlexNet
- VGG
- GoogLeNet
- ResNet
- DenseNet



小结 (Summary)

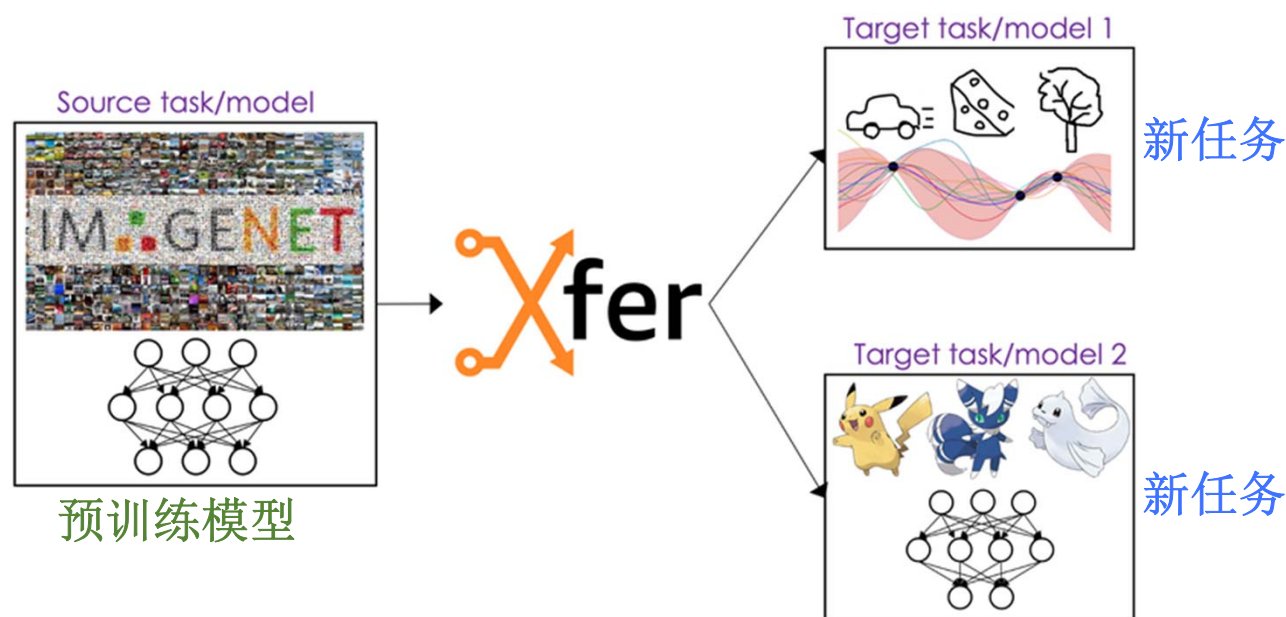
■ 卷积层的可视化

- 不同层级的卷积层学到的图像特征是分层的



■ CNN的预训练与微调

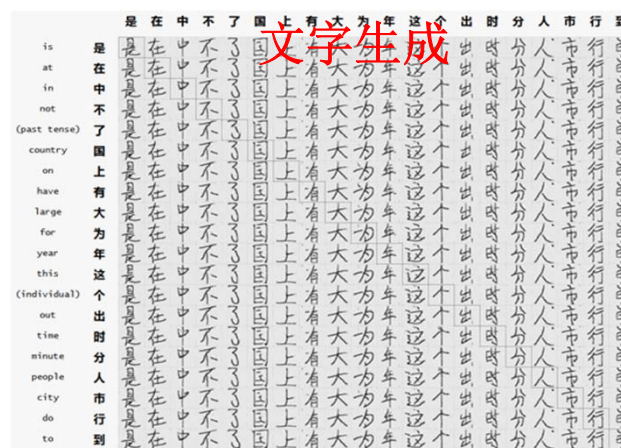
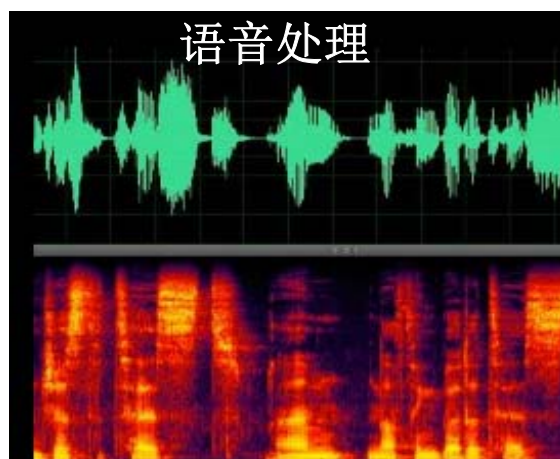
- 重复利用预训练模型，解决新任务
- 节省训练时间，提升新任务模型的收敛速度和性能
- 根据新任务的数据量大小、与预训练数据集的差异性，可选择对新任务模型中的哪些网络层进行冻结、微调或重新训练



小结 (Summary)

■ 卷积网络的应用

- 从自然语言处理到计算机视觉领域，CNN的身影无处不在



➤ An MIT Press book

“Deep Learning” by Ian Goodfellow *et al.*

➤ Online Open Courses

Convolutional Neural Networks for Visual Recognition, Stanford.

➤ Paper

- [1] Zeiler *et al.*, Visualizing and Understanding Convolutional Networks, ECCV2014.
- [2] Donahue et al, DeCAF: A Deep Convolutional Activation Feature for Generic Visual Recognition, ICML 2014.
- [3] Razavian et al, CNN Features Off-the-Shelf: An Astounding Baseline for Recognition, CVPR Workshops 2014.
- [4] Ren et al., Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks, NIPS 2015.
- [5] He et al, Rethinking ImageNet Pre-training, ICCV 2019.
- [6] Dumoulin V, *et al.*, . A guide to convolution arithmetic for deep learning[J]. arXiv preprint arXiv:1603.07285, 2016.

